EPIDEMIOLOGY AND MANAGEMENT OF RED ROT OF SUGARCANE CAUSED BY *Colletotrichum falcatum* WENT

BY

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in

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1992

Dedicated to
My Parents

CERTIFICATE -I

This is to certify that this dissertation entitled " Epidemiology and management of red rot of sugarcane caused by Colletotrichum falcatum Went" submitted for the degree of Ph.D., in the subject of Plant Pathology, of the Chaudhary Charan Singh Haryana Agricultural University, is a bonafide research work carried out by Shri Hawa Singh Saharan under my supervision and that no part of this dissertation has been submitted for any other degree.

The assistance and help received during the course of investigation has been fully acknowledged.

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CERTIFICATE-II

This is to certify that the dissertation entitled " Epidemiology and management of red rot of sugarcane caused by Colletotrichum falcatum Went" submitted by Shri Hawa Singh Saharan to the Chaudhary Charan Singh Haryana Agricutural University, in partial fulfilment of the requirements for the degree of Ph.D., in the subject of Plant Pathology, has been approved by the student's Advisory Committee after an oral examination on the same, in collaboration with an external examiner.

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Saharan

(Hawa Singh Saharan)

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CHAPTER-I

INTRODUCTION

Sugarcane is an important cash crop and the principal source of sugar in our country, covering the largest acreage among all the sugarcane growing countries of the world. It is grown in an area of 3.68 million hectares in India with the sugar production of 12.05 million tonnes during 1990-91. However, in Haryana, the area under sugarcane was 147 thousand hectares with the sugar production of 375 thousand tonnes during 1990-91 (Anonymous, 1991). The state has 11 sugar mills and a large number of gur and khandsari manufacturing units and thus sugarcane plays an important part in the economy of the state.

Among the major diseases of sugarcane, red rot caused by *Colletotrichum falcatum* Went (*Glomerella tucumanensis* (Speg.) Arx and Muller) is one of the earliest and has far reaching influence on the economics of sugar production in several countries. In India, red rot is the most serious disease of sugarcane, particularly in northern belt. Severe epiphytotics of red rot had occurred in this belt during 1938-40, 1946-47 and 1948-49, resulting in failure of some of our most popular varieties viz., Co 213, Co 312 and Co 453. The epidemic of 1938-40 in eastern Uttar Pradesh and northern Bihar had resulted in complete devastation of cane fields of the predominant commercial variety Co 213. A reduction of about 70,000 tonnes in sugar production of the country during 1938-39 was ascribed to red rot epidemic (Chona, 1956). In 1950-51, localised red rot epidemic affecting varieties Co 312, Co 313 and Co 421 was reported from Punjab (Chona and Srivastava, 1960). The outbreak of this disease has also been reported from Andhra Pradesh, Tamil Nadu, Pondicherry and Kerala (Prakasam and Venkatareddy, 1961; Lewin et al. 1976; Agnihotri and Singh, 1977; Nair et al., 1984).

The main cause of deterioration of several popular sugarcane varieties has been because of variability in the red rot pathogen. Beniwal et al. (1989) reported that red rot had been responsible for chasing out a number of commercial varieties and breakdown of a variety was attributed to the emergence of a new race. In general, breeding of red rot resistant early maturing high sugared varieties has not made much head way so far in our country. The possibility of obtaining resistant mutants through irradiation cannot be ruled out since this has been possible in certain other instances (Babu, 1969).

Detailed studies have been conducted on various aspects viz., pathogenic variability, mode and source of infection, host resistance and losses caused by this disease (Beniwal and Satyavir, 1991; Singh and Singh, 1989 and Singh et al., 1992). However, limited work has been carried out on epidemiology and disease management. Keeping all this in view, detailed studies with the following objectives were undertaken through the present investigations.

(i) To study disease epidemiology with particular reference to the role of environmental factors.

(ii) To study the role of nitrogen, soil amendments and fungicidal treatments in the red rot management.

(iii) To identify red rot resistant mutant.

Review of Literature

REVIEW OF LITERATURE

The main objective of this investigation was to study the epidemiology and management of red rot of sugarcane. The present brief reivew is, therefore, confined to a generalized picture of role of environmental factors, effect of different amendments, nitrogen levels and fungicidaal treatments on the development of red rot of sugarcane. In addition, the mutational pattern by gamma irradiation had also been emphasized. An attempt has been made in this context to highlight only those aspects of literature which have bearing on or relevance to the present study.

Red rot of sugarcane caused by Colletotrichum falcatum was first reported and described by Went in 1983 from Java (Indonesia) and proposed "red smut" as the common name for this disease. Within a decade, the occurrence of the disease was reported from several other countries, but it is regarded as a disease of major importance chiefly in the sub-tropical countries particularly the southern United States, India and Queensland (Chona, 1980). In India, the first outbreak of this disease was reported by Barber (1901) from Godaveri delta of Andhra Pradesh. Later, this disease was reported in epidemic form from northern Bihar and Bengal by Butler (1906) and he proposed " red rot " as common name for this disease which is now universally accepted.

A severe outbreak of this disease occurred in Jammu (Kashmir) in 1922 and in Pusa (Bihar) in 1932 (Chona, 1956). Red rot disease attracted special attention during the year 1938-40 when it appeared in epidemic form in northern Bihar and eastern Uttar Pradesh resulting in the failure of most popular variety Co 213 (Chona and Padwick, 1942; Chona, 1943). The disease again appeared in epidemic form during 1946-47 in several districts of Uttar Pradesh, this time resulting in the failure of popular variety Co 312 (Mathur, 1946). In 1948-49, it was again wide spread in eastern, mid-eastern and certain parts of western Uttar Pradesh affecting the variety Co 453 (Rafay, 1950). In 1950-51, localized red rot epidemic affecting Co 312, Co 313 and Co 421 was reported from Punjab (Chona and Srivastava, 1960). The disease has also been reported from Andhra Pradesh, Tamil Nadu, Pondicherry and Kerala causing serious losses in varieties Co 419, Co 658, Co 449 and Co 997 (Prakasam and Venkatareddy, 1961; Lewin et al., 1976; Agnihotri and Singh, 1977; Nair et al., 1984). In Haryana, red rot appeared in epidemic form during 1980-81 in our most popular variety Co 1148 (Satyavir et al., 1984) which was earlier found to be resistant to all the prevailing pathotypes in India (Alexander, 1979).

The ascigeral stage was first described by Spagazzini as Physalospora tucumanensis Speg. The species was not recognized as the perfect stage until Carvajal and Edgerton in 1943 reported it from Louisiana. Later Arx and Muller (1954) placed the perfect stage under genus Glomerella as G. tucumanensis (Speg.) Arx and Muller. This has gradually come to be accepted as the preferred name for the fungus. In India, the perithecial stage was not known to occur until Chona and Srivastava (1952) reported its occurrence in cultures. Subsequently, Chona and Bajaj (1953) found it in nature on sugarcane leaves at I.A.R.I, New Delhi. Singh et al. (1982)

also reported the occurrence of perfect stage of red rot pathogen on dried and semi-dried old leaves of sugarcanes from Gorakhpur in Uttar Pradesh.

Methods of inoculation and criteria for grading :

In India, the red rot of sugarcane is primarily a disease of standing canes. Therefore, standing canes are inoculated during rainy season and degree of attack is assessed 2-3 months after inoculation. Two methods of inoculation viz., plug and nodal are commonly used. Chona (1954) standardized red rot testing by plug method and used linear spread of infection for determining relative resistance of sugarcane varieties.

Chona (1950) tested the nodal resistance of sugarcane varieties by placing one ml of red rot inoculum into the space between leaves and stalk during the rainy season. Menon and Singh (1960) developed an inoculator which made plug method simpler and easier. Virk (1985) advocated the use of the modified hypodermic syringe for making puncture into internode and placement of inoculum.

Srinivasan and Bhat (1961) pointed out that linear spread of red rot in a variety was highly variable and red rot grading based on this character gave variable results. They suggested a new grading system (Symptomatic method) based on four characters, viz., drying of tops, lesion width, occurrence and nature of white spots and nodal transgression. Srinivasan (1962) developed a technique for screening young seedlings. In this method, six week old seedlings were inoculated by spraying the conidial suspension of C. falcatum and this method was found to be useful in eliminating more than 80 per cent of red rot susceptible seedlings at an early stage of growth. Singh et al. (1978) modified this method to use 5-6 months old plants (seedlings) instead of young seedlings.

Singh and Budhraja (1964) standardized nodal method (I.I.S.R. method) of inoculation and this was found to be more suitable for assessing field resistance of a variety. Sandhu and Babu (1967) observed that assessment of resistance to red rot based on width of lesion, occurrence of white spots, nodal transgression and drying of tops was more efficient than that of lesion length only. Kirtikar et al. (1972) adopted linear spread method and different varieties were classified into three different categories i.e. resistant (lesion length not exceeding 37.5 cm) moderately resistant, (lesion length 37.6 to 75.0 cm) and susceptible (lesion length above 75.0 cm).

Prasada Rao et al. (1978) observed that drying of tops is the most important field symptom for red rot which is the cumulative effect of lesion width (X_1), nature of white spots (X_2) and number of nodes transgressed (X_3). They assessed the relative influence of these factors on red rot and developed a mathematical model of discriminant function for grading varieties for resistance. The equation $Z = 3.5\ X_1 + 36.7\ X_2 + X_3$ took into consideration the relative values of three characters which were responsible for top drying.

Bergamin (1981) found that leaf inoculation method for evaluation of the resistance of sugarcane to C. falcatum was a little practical importance. Wang and Lee (1982) reported that drill tooth pick method was more convenient than the borer mycelia method for determining varietal resistance to red rot.

Singh et al. (1984) used nodal method for screening of sugarcane seedlings against red rot in seed beds under natural field conditions at Gorakhpur with range of temperature 25-33°C and relative humidity between 79 and 92 per cent. Alexander et al. (1986) used a new approach to improvement of red rot resistance in sugarcane. They made simultaneous tests on 200 genotypes using both natural infection and inoculation with C. falcatum in the four endemic zones of this disease (Karnal in Haryana, Chakia in northern Bihar, Tanuka in Andhra Pradesh and Nellikuppam in Tamil Nadu), and the results were discussed together with the advantages of testing the same set of genotype at different locations.

Behaviour of Colletotrichum falcatum in soil :

Opinions vary regarding survival of C. falcatum in soil (Chona and Nariani, 1952; Singh et al., 1977; Singh and Singh, 1983; Waraitch, 1983; Singh and Kumar, 1986). This is because some investigators used spores while others used infected tissues for studying the survival of red rot pathogen in soil. According to Chona and Nariani (1952) C. falcatum is capable of growing in the soil and produces abundent acervuli. They found that in natural soil or manured soil the fungus failed to survive, beyond three to four months. Singh et al. (1977) studied the survival using one cm long pieces of affected mid-ribs. These pieces were buried at different depths in soil and the survival of C. falcatum studied in two seasons, namely, winter and autumn. Periodic isolations from the affected material revealed that the pathogen survived for 63 days in winter and 34 days in autumn.

Waraitch (1983) found that C. falcatum survives in soil for 60 and 75 days after the incorporation of the debries under non-flooded and flooded conditions. Singh and Lal (1987) showed that the inoculum of C. falcatum remained viable in stagnant water for upto 60 days and for 7-8 months in infected tissues on the surface of damp soil. Evidence is presented that water plays an important role in the dissemination of inoculum and predisposes the host to penetration by the pathogen. Singh (1985) reported that disease incidence was reduced from 44% (debries burried just before planting) to 2.2% (90 days before planting). Infection was also observed in ratoon crop indicating that some inoculum persisted in soil even after uprooting the infected clumps from the plants crop. Waraitch (1985)confirmed the role of infested soil in disease development. Under Punjab conditions the pathogen survived in soil for 90 days in winter and 60 days in summer. Singh and Singh (1982) showed that the incubation of C. falcatum conidia in soil, using the buried slide technique, favoured germination and appressoria were formed 1-40 days later, thus indicating a long survival of the sugarcane pathogen in soil.

The infective propagules of C. falcatum in the soil may be spores, setae, chlamydospores, appressoria, ascospores or thickwalled hyphae (Singh and Singh, 1983; Singh et al., 1985). Singh et al. (1977) found that red rot infection in young seedlings occurred when 50 propagules of C. falcatum were present in a gram of soil. As the number of propagules increased in the soil, the red rot incidence also increased but 100% mortality of setts did not occur even when 0.5 million spores were initially mixed with one gram of soil.

Physiological study : Effect of temperature on growth and sporulation.

Edgerton and Moreland (1920) reported that the C. falcatum was able to grow at temperature ranging from 13-37.5°C with optimum near 27°C. Tims and Edgerton (1932) observed good growth of C. falcatum at 27-34°C. Abbott (1938) reported that minimum temperature for the growth of C. falcatum was 10°C, maximum slightly above 37.5°C and the optimum at 30-32.5°C. A temperature range of 25-28°C was found to be optimum for the growth of C. falcatum (Ramakrishnan, 1941). Chona and Hingorani (1950) conducted physiological studies on three isolates of *C. falcatum* and recorded optimum growth at a temperature of 30°C, very light at 10°C and nil at 40°C.

Ahmed and Divinagracia (1974) observed that *C. falcatum* grew at temperature from 10°C to 35°C with the optimum at 25°C - 30°C. Maximum sporulation occurred at 25°C but it was good at 20°C - 30°C. Light had no effect on growth but most sporulation occurred under continuous light and least in darkness. Khirbat (1980) reported the temperature range for growth of different isolates between 20°C and 30°C. The sporulation of all the isolates was maximum at 30°C.

Role of environmental factors on red rot development :

Only scanty information is available on the role of environmental factors on the development of red rot under field conditions.

Environmental factors affect both the spread and severity of red rot. Rana and Gupta (1968) reported good red rot development by spray inoculation method (nodal injury) during first fortnight of September under Shahjahanpur (Uttar Pradesh) conditions where average relative humidity was about 90 per cent and a temperature range of 24-33°C. Singh *et al.* (1984) reported good red rot development at temperature range of 25-30°C and relative humidity between 79 and 92 per cent. Influence of atmospheric temperature on red rot development (by plug method) was studied by Singh *et al.* (1988) as well as Beniwal and Satyavir (1991). They reported that the disease incidence was found to be highly responsive to variation in temperature. The red rot development was optimum between the mean temperature range of 29.4 - 31.0°C.

Role of mid rib-isolates :

The red rot pathogen of sugarcane, C. falcatum causes extensive lesion formation on mid ribs (Rajan and Singh, 1957). The fungus may remain confined to a small portion or may infect the entire mid rib, the intensity and incidence depending upon the susceptibility of the variety (Singh, 1966).

Butler (1906) obtained stalk rot from mid-rib isolates, while Abbott (1938) did not find mycelial connections between leaf and stalk lesions. Rafay and Singh (1957) reported severe pathogenicity with light coloured, mid-rib isolates, while dark coloured isolates of C. falcatum caused mild infection. On the other hand stalk inoculation trials with mid ribs isolates have shown that most of them are either non-pathogenic or weakly pathogenic (Agnihotri and Budhraja, 1972). According to Agnihotri and Budhraja (1974) the most mid-rib isolates of C. falcatum are weak pathogens and do not present a serious threat to cane cultivation. The races that cause mid rib lesions are markedly different from those causing stalk rot. Similarly, Sandhu et al. (1974) have concluded that mid rib isolates are less virulent than the stalk rot isolates. Lewin et al. (1978) studied the role of mid-ribs isolates of C. falcatum in causing red rot of sugarcane and found that the isolates from the mid-rib lesions did not cause stalk rot on even the susceptible varieties.

Effect of nitrogen on red rot development :

No work has been done on role of nitrogen fertilizer on red rot development. However, some physiological studies have been carried out to study effect of different forms of nitrogen on growth and sporulation of C. falcatum.

Ramakrishnan (1941) reported that the C. falcatum was able to metabolise a number of organic and inorganic sources of nitrogen of the several inorganic compounds namely protassium nitrate, calcum nitrate and sodium nitrate favoured good growth, while ammonium phosphate, potassium nitrate and sodium nitrate supported good sporulation. Manocha (1963) found that the addition of nitrogen boosted the sucrase or invertase production in both the dark and light coloured isolate.

Effect of amendment on red rot development :

Singh (1985) studied the effect of various amendment on red rot development in sugarcane. He demonstrated that soil treatment with organic and inorganic amendments increased development of red rot disease particularly when susceptible cultivar Co 312wasplanted, possibly it may be due to some stimulatory effect of soil amendment on the activity of the pathogen.

Fungicidal control of red rot :

Chemical therapy has been tried both against primary and secondary infections of red rot. Although several fungicides viz., bavistin, benomyl, vitavax, aretan and topsin have been found effective in vitro against C. falcatum, which have not given satisfactory control of the disease under field conditions (Singh and Singh, 1989).

Anzalone (1970) reported control of red rot in seed pieces with benomyl (100 or 200 ppm) when treated under air pressure of 10 lb p.s.i. for 15 minutes. Chand et al. (1974) reported that the treatment of setts of Co 312 with benomyl (0.25 per cent) and vitavax (0.05 per cent) increased germination over the control by 50 per cent and reduced the red rot incidence in the crop raised. Lewin et al. (1976) employed different organomercurials, antifungal antibiotics and systemic fungicides for controlling red rot. They reported that the dip treatment of setts agallol (0.05 per cent) for a period of 15 minutes gave good control of red rot. Waraitch (1983) recorded reduction in red rot when diseased seed pieces were dipped in bavistin solution (0.5 per cent) for one hour.

In a laboratory test with 14 fungicides against C. falcatum, Khirbat et al. (1984) demonstrated that in general the benzimidazoles performed best but plantvax and Hoe 6053 (pyracarbalid) also effectively inhibited spore germination.

Sharma et al. (1957) demonstrated that perenox was effective in checking the secondary spread of red rot. Kirtikar and Verma (1963) reported that blitox and dithane when sprayed three times between July and September reduced the spread of mid-rib lesions.

Induced mutations for red rot resistance :

By defination, a mutation is an inherited stable alteration to the genetic material (Brock, 1971). No assumption is made about the degree of experession of the genetic change, but mutation can only be recognized by alterations of the phenotype. The phenotype can be the whole organism or any part of it. As is now known, the genetic message encoded in the nucleotide base sequence. A mutation can, therefore, be an alteration to the nucleotide base sequence. A necleotide base change may or may not result in a recognizable change in the phenotype of the whole organisms but if it can be recognized as a change in DNA, it is by defination a mutation. A mutation, therefore, might have actually occured but may not be observed at that time. This will be actually so far characters governed by genes with small effects. This is, however, an important phenomenon with grreat consequence of evaluationary significance. It has been shown by different workers (Bhatia and Swaminathan, 1962; Brock, 1964) that as a result of irradiation the genotypic and phenotypic variability is increased with the difference in response depending on different varieties. According to Brock (1965), all characters respond with increased genetic variance. After discarding marphological mutants, the means of the irradiated populations as compared to control may not be altered. This is possible because mutations with plus and minus effects may occur equally frequently. However, means might sometimes deviate from control and may in fact be towards positive direction. This direction in the shift of means, according to Brock (1965), would depend upon previous selection, induced mutations do not follow a particular trend but would be random.

Natarajan (1964) suggested utilization of induced mutations for improving single trait of an established sugarcane variety like resistance to red rot or mosaic. Rao et al. (1966) obtained one plant each from the vegetative progeny of buds subjected to dosages of 500 R and 3000 R as resistant in variety Co 449. Some promising mutants of the variaties Co 527 and Co 663, have been obtained by Haq et al. (1970) following gamma radiation. Singh (1970) obtained a single mutant from irradiated setts which proved highly resistant to C. falcatum. Induced mutants with resistance to red rot have been obtained by Shah (1972) after radiation and chemical mutagenic treatment in the varieties Co 997, Co 449, Co 527 and Co 312. Nair (1973) was able to isolate Co 997-24-1, red rot resistant mutant which is similar to Co 997 for cane yield and sucrose content of the juice. Bari (1974) exposed buds of high yielding sugarcane varieties BL 4, BL 19, CoL 54 and L 116 to 2, 3 and 4 KR of gamma rays in an attempt to obtain plants resistant to red rot and mosaic. Haq et al. (1974) obtained nine mutants of Co 633 and six of Co 527 which showed different degrees of resistant to C. falcatum. The mutant

Co 527-85 was resistant in both field and laboratory. Haw et al. (1974) irradiated single-bud sets with 3 and 4 kR of gamma rays and tested for resistance to G. tucumanensis in the M_2 generation. A considerable increase in resistance was observed. Similarly, Jagathesan et al. (1974) irradiated single bud settsof varieties Co 419, CO 312, Co 527 and Co 602 with 3, 5, 6 and 8 kR of gamma rays and 5, 7.5 and 10 kR of X-rays. The variety Co 527 gave highest frequency of moderately resistant mutant in the M_2 generation followed by Co 312. Again, Jagathesan (1977) obtained mutations in a number of varieties in the Co series following mutagenic treatment of single bud setts particularly for dwarfness, flowerlessness, high sugar content glabrous leaves, increased growth rate and yield, and resistance to C. tucumanensis. Lo (1977) observed that F 153, F 158, Co 376 and S. spontaneum, EONS 32 were the most resistant clones to radiation demage.

Materials and methods

CHAPTER-III

MATERIALS AND METHODS

Multiplication and preparaion of inoculum :

Pathogenic isolate -RRK of C. falcatum (isolated from a cross at Regional Research Station, Uchani, Karnal, during 1986) was used for the present studies. The isolate was maintained on oat-meal agar slants at 4±1°C and renewed after every month.

The isolate was multiplied on oat meal agar medium in Roux bottles at 25±1°C. For inoculation, freshly sporulating 8-10 days old culture was taken. The spore mass was washed with sterile water and homogenised in warring blender for 2-3 minutes and spore suspension (2×10^4 conidia/ml) was prepared.

Method of inoculation :

Following three methods of inoculation were used during the course of study :

(i) Nodal mehod :

For nodal inoculation, 1 ml of spore suspension was taken and released in the space between leaf sheath and stalk at 3rd or 4th node from ground level with the help of a syringe. The inoculated canes were tagged.

(ii) Plug method :

In this method, second or third inter node of standing canes were inoculated using 0.2 ml spore suspension with the help of modified hypodermic syringe (Virk, 1985). Periodic observations on red rot development were recorded.

(iii) Soil inoculation method :

Red rot infected canes were collected, chopped in small pieces and mixed in furrows at the rate of 15 diseased canes per plot (7 x 2.25 m) just before planting in fungicidal experiment, whereas in soil amendmeent experiment, finely chopped 30 diseased canes were mixed in whole plot (7 x 2.25 m) one month before planting.

Recording of the disease :

(1) Nodal and soil inoculation methods :

Inoculated canes were observed for visual red rot symptoms. These included rind discolouration, sporulation over rind, yellowing and drying of canes. The per cent diseased plants were calculated.

(2) Plug method :

The canes were split open longitudinally at different periodic intervals and the following two methods were used for recording the disease:

(i) Lesion length method : Upward linear spread of red rot was measured from the point of inoculation and average linear spread was calculated (Chona, 1954).

(ii) Disease rating (Symptomatic method) : Disease rating was done on 0 - 9 scale. For calculating the disease rating, various criteria used by Srinivasan and Bhat (1961) with slight modifications were taken into consideration. For each criterian numerical ratings were given and disease rating was calculated by adding all the numerical values of observed canes and finally average disease index was worked out. The following criteria were used.

a) Nodal transgression :

0 = Lesion restricted within the inoculated internode.

1 = Lesion crossed one internode

2 = Lesion crossed two-three internodes.

3 = Lesion crossed more than three internodes.

b) Width of the lesion :

Width of the lesion was recorded in internode above the inoculated one.

1 = Lesion restricted in width i.e. covering 1/3rd or less width of the cane.

2 = Lesion spreading type i.e. covering 1/3rd to 2/3rd width of the cane.

3 = Lesion covering more than 2/3rd or entire width of the cane.

c) Occurrence of white spots :

0 = White spot absent.

1 = White spots restricted

2 = White spots fully developed.

d) Top drying :

0 = Top green

1 = Top yellow/dry.

Epidemiological Experiments :

(i) Role of environmental factors on red rot :

The experiment was conducted at sugarcane research area, Department of Plant Breeding, C.C.S.Haryana Agriculural University, Hisar and C.C.S. H.A.U. Regional Research Station, Uchani (Karnal) during the years 1989-90 and 1990-91. Three varieties namely Co 7717, CoL 29 and CoJ 64 were planted during both the years during second fortnight of March in blocks of 7 x 2.25 m with three replications.

Standing canes were inoculated by nodal method at fifteen days interval starting from the onset of monsoon. During the year 1989-90 there were three series of inoculations whereas during 1990-91, four series of inoculations were carried out at both the locations. Observations of diseased canes were recorded visually at 15 days interval after inoculation. The per cent diseased plants were calculated periodically in varieties Co 7717, CoL 29 and CoJ 64 symbolized as Y_1, Y_2 and Y_3, respectively.

The data on maximum temperature, minimum temperature, relative humidity morning, relative humidity evening and rainfall prevailed during the period of study were obtained from Department of Agro-meteorology, CCS HAU, Hisar and Central Soil Salinity Research Institute, Karnal.

The fifteen days mean of maximum temperature, minimum temperature, relative humidity morning, relative humidity evening and rainfall was calculated and symbolised as X_1, X_2, X_3, X_4 and X_5, respectively (Appendix 5). The data were analysed statistically by taking Y_1, Y_2 and Y_3 as dependent variable and X_1, X_2, X_3, X_4 and X_5 as independent variables using stepwise multiple regression analysis (SMRA) (Singh et al., 1990).

Progress of red rot was plotted on the graph by taking time period on X axis and $\log_e$ (1/1-x) transformed value and per cent disease incidence on Y axis (Appendix 1.1 - 4.4).

(ii) Role of mid-rib isolats of C. falcatum in red rot development :

Mid-rib samples showing red rot symptoms were collected from different varieties and localities of Haryana during the month of July, 1990. These were cut into small bits, surface sterilized with 0.1 per cent mercuric chloride and placed on oat-meal agar slants at 25 ± 1°C. After the identification, single spore isolates were made to purify the cultures and these were maintained as per method described earlier. The mid-rib isolates collected from different varieties and localities are given be below :

Mid-rib isolate No.	Variety	Locality
MR_1	CoJ 64	Yamunanagar
MR_2	CoJ 64	Karnal
MR_3	Co 7314	Yamunanagar
MR_4	Co 7314	Karnal
MR_5	CoH 89	Karnal
MR_6	Co 1158	Yamunagar
MR_7	CoH 88	Karnal
MR_8	CoH 3	Karnal
MR_9	Co 7717	Karnal
MR_{10}	Co 7717	Hisar

Pathogenic behaviour of mid-rib isolates was studied by plug method of inoculation on varieties Co 7717, CoJ 64, CoL 29 during third week of September, 1990 and observations were recorded after two months of inoculation.

Red-rot Management :

(i) Effect of nitrogen on red rot development :

Effect of different doses of nitrogen on the development of red rot was studied. Five levels of nitrogen in the form of urea, i.e. 0, 75, 150, 225 and 300 kg N/ha were applied in two split doses. Three varieties namely CoJ 64, Co 7717 and CoS 767 were planted at experimental area of sugarcane, Department of Plant Breeding, C.C.S.H.A.U.. Hisar and C.C.S.H.A.U. Regional Research Station, Uchani (Karnal) during the year 1990-91. The plot size under this experiment was 8 x 3 m with three replications. Canes were inoculated by plug method during third week of August, 1990. Observations on disease rating were recorded one month after inoculation by split opening the 30 canes in each replication of every treatment.

(ii) Effect of soil amendments on red-rot development:

The study was conducted to determine the effect of various amendments on the soil inoculum of *C. falcatum* under field conditions and their role in disease development.

The experiment was conducted in randomized block design with a plot size of 7 x 2.25 m with three replication at Sugarcane Research area of Department of Plant Breeding, C.C.S.H.A.U., Hisar during 1990-91. Red rot infected canes were collected chopped in small pieces and mixed in the soil at 10 cm depth at the rate of 30 diseased canes per plot.

Later, these plots were amended with different soil amendments, namely sugarcane baggase @ 30 and 40 q/ha and press mud @ 40 and 50 q/ha in the top 15 cm soil. After 20 days of amendment, planting was done. Inorganic salts, viz., zinc sulphate and ferrous sulphate were mixed at the time of planting in furrows @ 15 and 25kg/ha each. Healthy three budded setts of susceptible variety CoL 29 were planted. Disease incidence on clump and cane basis in per cent were recorded. Data were analysed after subjected to angular transformation.

(iii) Efficacy of Bavistin in controlling red rot :

The role of Bavistin, a systematic fungicide in controlling the disease was investigated. The experiment was conducted in randomized block design, with a plot size of 7 x 2.25 m with three replications at Sugarcane Research area of Department of Plant Breeding, C.C.S.H.A.U., Hisar during 1990-91. Red rot infected canes were collected, chopped in small pieces and mixed in furrows at the rate of 15 diseased canes per plot before planting. Three budded healthy setts of a highly susceptible cultivar CoL 29 were treated with Bavistin @.25% for one and two hours alone and in combination with one spray and two sprays of Bavistin @ 0.1%. The first and second foliar sprays with Bavistin was done on August 15 and November 4, 1990, respectively. Disease incidence (per cent) on clump and cane basis were recorded and analysed after subjected to angular transformation.

Identification of red rot resistant mutant :

The experimental material for this study consisted of sugarcane varieties Co 8334 and CoH 56. Single budded setts of these varieties were subjected to mutagenic treatment during the month of February, 1989.

Dormant single budded setts were given gamma radiation exposures of 4.0 kilo roentgen (kR) from CO_{60} source at gamma chamber of Department of Genetics, C.C.S.H.A.U., Hisar. While considering the amount of dose to be used, previous reported work on sugarcane (Vijaylaxmi and Rao, 1960; Rao et al., 1966; Hans, 1973) was kept in mind.

Irradiated single budded setts alongwith controls were planted in raised nursery buds. Survival count was recorded after 45 days. Subsequently these plants were transplanted in the field at a distance of 1 x 0.30 m. All the canes in different clones were inoculated in first week of September, 1989 by nodal method of inoculation. Visual observation on red rot was recorded. Apparently healthy looking clones were selected from both the varieties and were planted in single rows during the next season alongwith controls. Half of the millable canes in each clone were again inoculated by nodal method with the onset of monsson and observed for red rot reaction.

Statistical Analysis :

Data collected during the present study were statistically analysed by applying randomized block design except epidemiological experiment. To develop the predictive models for red rot development disease incidence was regressed with weather parameters by applying step-wise multiple regression analysis (Singh et al., 1990).

Results

and

Discussion

CHAPTER-IV

RESULTS AND DISCUSSION

Epidemiology of red rot of sugarcane :

Role of environmental factors on red rot development :

Attempts were made to develop predictive models for the red rot of sugarcane by conducting the experiments at Hisar and Karnal during the years 1989-90 and 1990-91. Standing canes of varieties Co 7717, CoL 29 and CoJ 64 were inoculated by nodal method at 15 days interval starting from the onset of monsoon. There were three and four series of inoculations at both the locations during 1989-90 and 1990-91. Weather data (Appendix 5) were regressed with disease incidence (Appendix 1.1 -4.4) applying stepwise multiple regression analysis (SMRA) and best fitting equations were generated following SMRA technique (Singh et al., 1990).

Table 1 : The best fitting linear models for predicting incidence of red rot

Sr.No.		Regression equation	R^2
1	Y_1	$= 118.12 - 4.52X_2$	0.87
2	Y_2	$= 231.03 - 8.58X_2$	0.86
3	Y_3	$= 50.43 - 1.84X_2$	0.82

Location - Hisar, Date of inoculation - 15.7.1989

ANOVA - Table 1 for regression of Y_1 on X_2

Source	D.F.	SS	MS	F value	Prob.
Regression	1	7803.98	7803.98	131.51	5.552E-10
Residual	19	1127.47	59.34		
Total	20	8931.45			

ANOVA - Table 1 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	28117.68	28117.68	117.41	1.424E-09
Residual	19	4550.36	239.49		
Total	20	32668.04			

ANOVA - Table 1 for regression of Y_3 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	1287.90	1287.90	89.07	1.326E-08
Residual	19	274.73	14.46		
Total	20	1562.63			

Under Hisar conditions during 1989-90, when the canes were inoculated on July 15, 1989, the best fitting models developed were eq. 1, 2 and 3 for Y_1, Y_2 and Y_3, respectively. The minimum temperature was the most important parameter which was negatively correlated during the succeeding fortnight. The R^2 values of 0.87, 0.86 and 0.82 was indicating a very good fit yet 13, 14 and 18 per cent of the variability of red rot incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 1 and 2).

Table 2 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	0.58	1.00						
X_3	0.63	0.19	1.00					
X_4	-0.09	0.74	0.77	1.00				
X_5	-0.46	0.32	0.93	0.82	1.00			
Y_1	-0.40	-0.94	-0.26	-0.78	-0.32	1.00		
Y_2	-0.36	-0.93	-0.26	-0.77	-0.34	0.96	1.00	
Y_3	-0.42	-0.91	-0.22	-0.71	-0.35	0.85	0.94	1.00

Critical value [2 Tail, .05) = ± 0.43

Location - Hisar, Date of inoculation - 15.7.1989.

Table 3 : The best fitting linear models for predicting incidence of red rot.

Sr.No.		Regression equation	R^2
4	Y_1	$= 54.86 - 2.09X_2$	0.63
5	Y_2	$= 68.35 - 2.63X_2$	0.80
6	Y_3	$= 32.24 - 1.25X_2$	0.81

Location - Hisar, Date of inoculation - 31.7.1989

ANOVA - Table 3 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	Prob
Regression	1	1365.85	1365.85	26.94	8.934E-05
Residual	16	811.21	50.70		
Total	17	2177.05			

ANOVA-Table 3 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	2158.33	2158.33	63.45	5.864E-07
Residual	16	544.25	34.02		
Total	17	2702.58			

ANOVA -Table 3 for regression of Y_3 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	490.44	490.44	68.47	3.569E-07
Residual	16	114.60	7.16		
Total	17	605.04			

Under Hisar conditions during 1989-90 when the canes were inoculated on July 31, 1989, the best fitting models developed were eq. 4, 5 and 6 for Y_1, Y_2 and Y_3, respectively. The minimum temperature was the most important parameter which was negatively correlated during the succeeding fortnight. The R^2 value of 0.63, 0.80 and 0.81 was indicating a good fit yet 37, 20 and 19 per cent of the variability of red rot incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 3 and 4).

Table 4 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	0.49	1.00						
X_3	-0.63	0.28	1.00					
X_4	-0.11	0.80	0.78	1.00				
X_5	-0.44	0.43	0.93	0.83	1.00			
Y_1	-0.22	-0.79	-0.28	-0.72	-0.41	1.00		
Y_2	-0.24	-0.89	-0.35	-0.82	-0.46	0.73	1.00	
Y_3	-0.24	-0.90	-0.37	-0.83	-0.47	0.87	0.92	1.00

Critical value (2-Tail, .05) = ± 0.47 , Location-Hisar, Date of inculation 31.7.89

Table 5 : The best fitting linear models for predicting incidence of red rot

Sr.No.		Regression equation	R^2
7	Y_1	$= 40.31 - 1.60X_2$	0.87
8	Y_2	$= 15.51 - 0.26X_4$	0.83
9	Y_3	$= 8.13 - 0.14X_4$	0.76

Location - Hisar, Date of inoculation - 15.8.1989

ANOVA-Table 5 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	627.20	627.20	86.94	4.028E-07
Residual	13	93.79	7.22		
Total	14	720.99			

ANOVA-Table 5 for regression of Y_2 on X_4

Source	DF	SS	MS	F value	Prob.
Regression	1	230.66	230.66	63.66	2.302E-06
Residual	13	47.10	3.62		
Total	14	277.77			

ANOVA-Table 5 for regression of Y_3 on Y_4

Source	DF	SS	MS	F value	Prob.
Regression	1	64.38	64.38	40.90	2.362E-05
Residual	13	20.47	1.57		
Total	14	84.84			

Under Hisar conditions during 1989-90, when the canes were inoculated on August 15, 1989, the best fitting model developed was eq. 7 for Y_1. The minimum temperature was the most important parameter which was negatively correlated during the succeeding fortnight. The R^2 value of 0.87 was indicating a good fit yet 13 per cent of the variability of red rot of sugarcane incidence remained uncounted. The best fitting models developed were eq. 8 and 9 for Y_2 and Y_3, respectively, where the relative humidity evening was the most important parameter, which was negatively correlated during the succeeding fortnight. The R^2 value of 0.83 and 0.76 was indicating a good fit yet 17 and 24 per cent of the variability of red rot incidence in Y_2 and Y_3, respectively remained uncounted (Table 5 and 6).

Table 6 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	0.30	1.00						
X_3	-0.54	0.56	1.00					

Contd...........

Table 6 Contd............

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_4	-0.18	0.88	0.87	1.00				
X_5	-0.41	0.61	0.94	0.86	1.00			
Y_1	0.78	-0.93	0.50	-0.86	-0.60	1.00		
Y_2	0.80	-0.86	-0.64	-0.91	-0.68	0.93	1.00	
Y_3	0.12	-0.81	-0.59	-0.87	-0.64	0.94	0.95	1.00

Critical value (2-Tail; .05) = ± 0.51

Location - Hisar, Date of inoculation - 15.8.1989

Table 7 : The best fitting linear models for predicting incidence of red rot

Sr.No.			Regression equation	R^2
10	Y_1	=	$286.42 - 10.27X_2$	0.74
11	Y_2	=	$249.05 - 8.94X_2$	0.86
12	Y_3	=	$57.65 - 2.10X_2$	0.76

Location - Karnal, Date of inoculation - 20.7.1989

ANOVA-Table 7 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	18638.48	18638.48	45.67	4.600E-06
Residual	16	6530.18	408.14		
Total	17	25168.66			

ANOVA-Table 7 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	14109.52	14109.52	97.57	3.258E-08
Residual	16	2313.66	144.60		
Total	17	16423.18			

ANOVA-Table 7 for regression of Y_3 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	779.30	779.30	50.59	2.462E-06
Residual	16	246.49	15.41		
Total	17	1025.79			

Under Uchani (Karnal) conditions during 1989-90, when the canes were inoculated on July 20, 1989, the best fitting models developed were eq. 10, 11 and 12 for Y_1, Y_2 and Y_3, respectively. The minimum temperature was the most important parameter which was negatively correlated during the succeeding fortnight. The R^2 value of 0.74, 0.86 and 0.76 was indicating a good fit yet 26, 14 and 24 per cent of the variability of red rod incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 7 and 8).

Table 8 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	-0.37	1.00						
X_3	-0.64	0.69	1.00					
X_4	-0.73	0.90	0.86	1.00				
X_5	-0.94	0.52	0.55	0.78	1.00			
Y_1	0.45	-0.86	-0.47	-0.79	-0.68	1.00		
Y_2	0.18	-0.93	-0.44	-0.73	-0.43	0.92	1.00	
Y_3	0.17	-0.87	-0.39	-0.68	-0.41	0.94	0.95	1.00

Critical value (2 Tail, .05) = ± 0.47
Location - Karnal, Date of inoculation - 20.7.1989

Table 9 : The best fitting linear models for predicting incidence of red rot

Sr.No.	Regression equation	R^2
13	$Y_1 = 53.13 - 1.97X_2$	0.90
14	$Y_2 = 179.16 - 6.63X_2$	0.97
15	$Y_3 = 23.82 - 0.91X_2$	0.86

Location - Karnal, Date of inoculation - 5.8.1989

ANOVA-Table 9 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	576.31	576.31	112.58	9.011E-08
Residual	13	66.55	5.12		
Total	14	642.85			

ANOVA-Table 9 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	6531.20	6531.20	416.05	2.960E-11
Residual	13	204.07	15.70		
Total	14	6735.27			

ANOVA-Table 9 for regression of Y_3 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	122.20	122.20	77.67	7.632E-07
Residual	13	20.45	1.57		
Total	14	142.66			

Under Uchani (Karnal) conditions during 1989-90 when the canes were inoculated on August 5, 1989, the best fitting models developed were eq. 13, 14 and 15 for Y_1, Y_2 and Y_3, respectively. The minimum temperature was the most important parameter which was negatively correlected during the succeeding fortnight. The R^2 value of 0.90, 0.97 and 0.86 was indicating a very good fit yet 10, 3 and 14 per cent of the variability of red rot incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 9 and 10).

Table 10 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	-0.41	1.00						
X_3	-0.66	0.87	1.00					
X_4	-0.74	0.91	0.94	1.00				
X_5	-0.99	0.45	0.65	0.77	1.00			
Y_1	0.39	-0.95	-0.83	-0.87	-0.46	1.00		
Y_2	0.42	-0.98	-0.90	-0.91	-0.47	0.96	1.00	
Y_3	0.49	-0.93	-0.79	-0.89	-0.56	0.88	0.93	1.00

Critical value (2 Tail; .05) = ± .51

Location -- Karnal, Date of inoculation -- 5.8.1989.

Table 11 : The best fitting linear models for predicting incidence of red rot

Sr.No.		Regression equation	R^2
16	Y_1	$= 28.55 - 0.32X_4$	0.97
17	Y_2	$= 24.56 - 0.25X_4$	0.91
18	Y_3	$= 9,94 - 0.12X_4$	0.87

Location -- Hisar, Date of inoculation -- 20.8.1989

ANOVA-Table 11 for regression of Y_1 on X_4

Source	DF	SS	MS	F value	Prob.
Regression	1	302.43	302.43	328.55	5.603E-09
Residual	10	9.21	0.92		
Total	11	311.63			

ANOVA-Table 11 for regression of Y_2 on X_4

Source	DF	SS	MS	F value	R^2 value
Regression	1	194.86	194.86	97.07	1.821E-06
Residual	10	20.07	2.01		
Total	11	214.93			

ANOVA - Table 11 for regression of Y_3 on X_4

Source	DF	SS	MS	F value	R^2 value
Regression	1	46.20	46.20	69.62	8.132E-06
Residual	10	6.64	0.66		
Total	11	52.84			

Under Uchani (Karnal) conditions during 1989-90, when the canes were inoculated on August 20, 1989, the best fitting models developed were eq. 16, 17 and 18 for Y_1, Y_2 and Y_3, respectively. The relative humidity evening was the most important parameter which was negatively correlated during the succeeding fortnight. The R^2 value of 0.97, 0.91 and 0.87 was indicating a very good fit yet 3, 9 and 13 per cent of the variability of red rot incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 11 and 12).

Table 12 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	-0.59	1.00						
X_3	-0.75	0.89	1.00					
X_4	-0.85	0.93	0.94	1.00				
X_5	-0.10	0.54	0.70	0.83	1.00			
Y_1	0.77	-.096	-0.91	-0.99	-0.75	1.00		

Contd..........

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
Y_2	0.73	-0.94	-0.93	-0.95	-0.70	0.94	1.00	
Y_3	0.72	-0.93	-0.84	-0.94	-0.71	0.95	0.91	1.00

Critical value (2 Tail, .05) = ± 0.57

Location --- Karnal, Date of inoculation -- 20.8.1989.

Table 13 : The best fitting linear models for predicting incidence of red rot

Sr.No.		Regression equation	R^2
19	Y_1	$= 67.77 - 0.74X_4$	049
20	Y_2	$= 36.86 - 1.31X_2$	0.76

Location -- Hisar, Date of inoculation -- 3.7.1990.

ANOVA- Table 13 for regression of Y_1 on X_4

Source	DF	SS	MS	F value	Prob.
Regression	1	2297.26	2297.26	20.68	3.205E-04
Residual	22	2443.46	111.07		
Total	23	4740.72			

ANOVA-Table 13 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	772.62	772.62	70.69	1.133E-07
Residual	22	240.55	10.93		
Total	23	1013.17			

Under Hisar conditions during 1990-91 when the canes were inoculated on July 3, 1990, the best fitting models developed were eq. 19 and 20 for Y_1 and Y_2, respectively. The relative humidity evening was the most important parameter for Y_1 which was negatively correlated during the succeeding fortnight. The R^2 value of 0.49 was indicating a good fit yet 51 per cent of the variability of red rot incidence in Y_1 remained uncounted whereas, minimum temperature showed negative correlation with Y_2. The R^2 value of 0.76 was indicating a very good fit yet 24 per cent of the variability of red rot incidence in Y_2 remained uncounted (Table 13 and 14).

Table 14 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2
X_1	1.00						
X_2	0.89	1.00					
X_3	0.66	0.76	1.00				
X_4	0.77	0.96	0.78	1.00			

Contd............

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2
X_5	0.14	0.46	0.42	0.63	1.00		
Y_1	-0.56	-0.68	-0.54	-0.70	-0.32	1.00	
Y_2	-0.79	-0.87	-0.75	-0.85	-0.36	0.82	1.00

Critical value (2 Tail, .05) = ± 0.42

Location -- Hisar, Date of inoculation -- 3.7.1990

Table 15 : The best fitting linear models for predicting incidence of red rot.

Sr.No.		Regression equation	R^2
21	Y_1	$= 37.65 - 1.28X_2$	0.72
22	Y_2	$= 31.74 - 1.23X_4$	0.96

Location - Hisar, Date of inoculation -- 19.7.90

ANOVA - Table 15 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	711.18	711.18	49.48	1.072E-06
Residual	19	273.12	14.37		
Total	20	984.30			

ANOVA - Table 15 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	654.62	654.62	464.17	.000E -0.00
Residual	19	26.80	1.41		
Total	20	681.42			

Table 16 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2
X_1	1.00						
X_2	0.91	1.00					
X_3	0.66	0.79	1.00				
X_4	0.77	0.96	0.79	1.00			
X_5	0.14	0.45	0.43	0.62	1.00		
Y_1	-0.81	-0.85	-0.70	-0.84	-0.32	1.00	
Y_2	-0.90	-0.98	-0.81	-0.94	-0.42	0.87	1.00

Critical value (2 Tail, .05) = ± 0.43

Location -- Hisar, Date of inoculation 19.7.1990

Under Hisar conditions during 1990-91 when the canes were inoculated on July 19, 1990; the best fitting models developed were eq. 21 and 22 for Y_1 and Y_2, respectively. The minimum temperature was the most important parameter which was negatively correlated during the succeeding fortnight. The R^2 value of 0.72 and 0.96 was indicating a very good fit yet 18 and 4 per cent of the variability of red rot incidence in Y_1 and Y_2, respectively remained uncounted (Table 15 and 16).

Table 17 : The best fitting linear models for predicting incidence of red rot

Sr.No.		Regression equation	R^2
23	Y_1	$= 11.18 - 0.39X_2$	0.53

Location -- Hisar, Date of inoculation - 2.8.1990.

ANOVA - Table 17 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	R^2
Regression	1	58.20	58.20	18.13	6.012E-04
Residual	16	51.36	3.21		
Total	17	109.56			

Under Hisar conditions during 1990-91 when the canes were inoculated on August 2, 1990, the best fitting linear model developed was eq. 23 for Y_1.

The minimum temperature was the most important parameter which was negatively correlated during the succeeding fortnight. The R^2 value of 0.53 indicating a good fit yet 47 per cent of variability of red rot incidence remained uncounted (Table 17 and 18).

Table 18 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2
X_1	1.00						
X_2	0.90	1.00					
X_3	0.65	0.79	1.00				
X_4	0.74	0.96	0.81	1.00			
X_5	0.09	0.41	0.42	0.61	1.00		
Y_1	-0.68	-0.73	-0.63	-0.69	-0.24	1.00	
Y_2	-0.40	-0.47	-0.38	-0.46	-0.30	0.04	1.00

Critical value (2 Tail, .05) = ± 0.47

Location - Hisar, Date of inoculation - 3.8.1990

Table 19 : The best fitting linear models for predicting incidence of red rot

Sr.No.			Regression equation	R^2
24	Y_1	=	$4.57 - 0.18X_2$	0.35
25	Y_2	=	$1.74 - 0.03X_4$	0.68

Location - Hisar, Date of inoculation - 17.8.1990.

ANOVA- Table 19 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	10.52	10.52	6.85	0.0213
Residual	13	19.96	1.54		
Total	14	30.47			

ANOVA - Table 19 for regression of Y_2 on X_4

Source	DF	SS	MS	F value	Prob.
Regression	1	2.44	2.44	11.27	5.149E-03
Residual	13	2.82	0.22		
Total	14	5.26			

Under Hisar conditions during 1990-91 when the canes were inoculated on August 17, 1990, the best fitting models developed were eq. 24 for Y_1 and Y_2, respectively. The minimum temperature for Y_1 and relative humidity evening for Y_2 were the most important parameter which were negatively correlated during the succeeding fortnight. The R^2 value of 0.35 and 0.68 was indicating a good fit yet 65 and 32 per cent of the variability of red rot incidence in Y_1 and Y_2, respectively remained uncounted (Table 19 and 20).

Table 20 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2
X_1	1.00						
X_2	0.91	1.00					
X_3	0.72	0.77	1.00				
X_4	0.74	0.95	0.77	1.00			
X_5	0.06	0.41	0.45	0.63	1.00		
Y_1	-0.57	-0.59	-0.51	-0.54	-0.20	1.00	
Y_2	-0.56	-0.68	-0.46	-0.68	-0.31	0.01	1.00

Critical value (2 Tail, .05) = ± 0.51

Location - Hisar, Date of inoculation — 17.8.1990

Table 21 : The best fitting linear models for predicting incidence of red rot.

Sr.No.	Regression equation	R^2
26	$Y_1 = 157.84 - 1.29X_4$	0.56
27	$Y_2 = 129.68 - 4.27X_2$	0.76
28	$Y_3 = 21.70 - 0.77X_2$	0.56

Location -- Karnal, Date of inoculation -- 5.7.1990

ANOVA - Table 21 for regression of Y_1 on X_4

Source	DF	SS	MS	F value	Prob.
Regression	1	13793.18	13793.18	12.60	1.383E-03
Residual	28	30641.54	1094.34		
Total	29	44434.71			

ANOVA - Table 21 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	19872.53	19872.53	87.48	4.123E-10
Residual	28	6360.53	227.16		
Total	29	26233.05			

ANOVA- Table 21 for regression of Y_3 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	638.75	638.75	35.80	1.914E-06
Residual	28	499.56	17.84		
Total	29	1138.31			

Under Uchani (Karnal) conditions during 1990-91, when the canes were inoculated on July 5, 1990; the best fitting modelsdeveloped were eq. 26, 27 and 28 for Y_1, Y_2 and Y_3, respectively. The relative humidity evening for Y_1 and minimum temperature for Y_2 and Y_3 were the most important parameters which were negatively correlated during the succeeding fortnight. The R^2 value of 0.56, 0.76 and 0.56 was indicating a good fit yet 34, 24 and 34 per cent of the variability of red rot incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 21 and 22).

Table 22 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	0.89	1.00						
X_3	0.34	0.52	1.00					
X_4	0.56	0.87	0.60	1.00				
X_5	0.48	0.63	0.70	0.65	1.00			
Y_1	-0.43	-0.54	-0.15	-0.56	-0.49	1.00		
Y_2	-0.73	-0.87	-0.38	-0.83	0. 61	0.84	1.00	
Y_3	-0.60	-0.75	-0.43	-0.74	-0.58	0.59	0.79	1.00

Critical value (2 Tail, .05) = ± 0.36

Location -- Karnal, Date of inoculation -- 5.7.1990.

Table 23: The best fitting linear models for predicting incidence of red rot

Sr.No.	Regression equation	R^2
29	$Y_1 = 189.67 - 2.04X_4$	0.68
30	$Y_2 = 47.90 - 1.59X_2$	0.65
31	$Y_3 = 8.90 - 0.30X_2$	0.52

Location - Karnal, Date of inoculation -- 21.7.1990

ANOVA - Table 23 for regression of Y_1 on X_4

Source	DF	SS	MS	F value	Prob.
Regression	1	31185.41	31185.41	53.54	1.149E-07
Residual	25	14561.88	582.48		
Total	26	45747.29			

ANOVA- Table 23 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	2505.32	2505.32	47.21	3.344E-07
Residual	25	1326.59	53.06		
Total	26	3831.91			

ANOVA - Table 23 for regression of Y_3 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	88.89	88.89	27.11	2.183E-05
Residual	25	81.96	3.28		
Total	26	170.85			

Under Uchani (Karnal) conditions during 1990-91 when the canes were inoculated on July 21, 1990, the best fitting models developed were eq. 29, 30 and 31 for Y_1, Y_2 and Y_3, respectively. The relative humidity evening for Y_1 and minimum temperature for Y_2 and Y_3 were the most important parameters which were negatively correlated during the succeeding fortnight. The R^2 value of 0.68, 0.65 and 0.52 was indicating a good fit yet 32, 35 and 48 per cent of the variability of red rot of sugarcane incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 23 and 24).

Table 24 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	0.88	1.00						
X_3	0.43	0.66	1.00					
X_4	0.52	0.86	0.74	1.00				

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_5	0.50	0.67	0.72	0.69	1.00			
Y_1	-0.61	-0.80	-0.65	-0.83	-0.78	1.00		
Y_2	-0.67	-0.81	-0.54	-0.75	-0.62	0.80	1.00	
Y_3	-0.59	-0.72	-0.50	-0.70	-0.59	0.85	0.78	1.00

Critical value (2 Tail, .05) = ± 0.38

Location -- Karnal, Date of inoculation - 21.7.1990.

Table 25 : The best fitting linear models for predicting incidence of red rot

Sr.No.		Regression equation	R^2
32	Y_1	$= 144.70 - 5.35X_2$	0.90
33	Y_2	$= 26.30 - 0.93X_2$	0.69
34	Y_3	$= 5.72 - 0.21X_2$	0.57

Location -- Karnal, Date of inoculation - 6.8.1990

ANOVA - Table 25 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	25077.06	25077.06	198.24	1.720E-12
Residual	22	2782.93	126.50		
Total	23	27859.98			

ANOVA - Table 25 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	761.97	761.97	48.65	5.314E-07
Residual	22	344.57	15.66		
Total	23	1106.54			

ANOVA - Table 25 for regression for Y_3 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	39.56	39.56	29.59	1.829E-05
Residual	22	29.41	1.34		
Total	23	68.97			

Under Uchani (Karnal) conditions during 1990-91 when the canes were inoculated on August 6, 1990, the best fitting models developed were eq. 32, 33 and 34 for Y_1, Y_2 and Y_3, respectively. The minimum temperature was the most important parameters which was negatively correlated during the succeeding fortnight. The R^2 value of 0.90, 0.69 and 0.57 was indicating a good fit yet 13, 14 and 18 per cent of the variability of red rot incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 25 and 26).

Table 26 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	0.87	1.00						
X_3	0.35	0.60	1.00					
X_4	0.47	0.84	0.71	1.00				
X_5	0.41	0.67	0.68	0.74	1.00			
Y_1	-0.76	-0.95	-0.64	-0.87	-0.65	1.00		
Y_2	-0.68	-0.83	-0.45	-0.74	-0.53	0.72	1.00	
Y_3	-0.58	-0.76	-0.49	-0.72	-0.65	0.71	0.85	1.00

Critical value (2 Tail, .05) = ± 0.40

Location - Karnal, Date of inoculation - 6.8.1990.

Table 27 : Correlation of red rot with weather parameters

	X_1	X_2	X_3	X_4	X_5	Y_1	Y_2	Y_3
X_1	1.00							
X_2	0.87	1.00						
X_3	0.32	0.58	1.00					
X_4	0.43	0.81	0.72	1.00				
X_5	0.43	0.75	0.71	0.88	1.00			
Y_1	-0.87	-0.98	-0.49	-0.77	-0.70	1.00		
Y_2	-0.82	-0.95	-0.50	-0.77	-0.69	0.96	1.00	
Y_3	-0.59	-0.66	-0.46	-0.52	-0.56	0.64	0.53	1.00

Critical value (2 Tail, .05) = ± 0.43

Location - Karnal, Date of inoculation - 21.8.1990.

Table 28 : The best fitting linear models for predicting incidence of red rot.

Sr.No.		Regression equation	R^2
35	Y_1	$= 148.01 - 5.82X_2$	0.96
36	Y_2	$= 20.75 - 0.83X_2$	0.90
37	Y_3	$= 2.94 - 0.11X_2$	0.44

Location - Karnal, Date of inoculation - 21.8.1990.

ANOVA - Table 28 for regression of Y_1 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	24984.96	24984.96	481.39	0.00
Residual	19	986.14	51.90		
Total	20	25971.10			

ANOVA - Table 28 for regression of Y_2 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	499.28	499.28	176.63	4.533E-11
Residual	19	53.71	2.83		
Total	20	552.99			

ANOVA - Table 28 for regression of Y_3 on X_2

Source	DF	SS	MS	F value	Prob.
Regression	1	9.04	9.04	14.75	1.102E-03
Residual	19	11.64	0.61		
Total	20	20.68			

Under Uchani (Karnal) conditions during 1990-91 when the canes were inoculated on August 21, 1990, the best fitting models developed were eq. 35, 36 and 37 for Y_1, Y_2 and Y_3, respectively. The minimum temperature was the most important parameter which was negatively correlated during the succeeding fortnight. The R^2 value of 0.96, 0.90 and 0.44 was indicating a very good fit yet 4, 10 and 64 per cent of the variability of red rot incidence in Y_1, Y_2 and Y_3, respectively remained uncounted (Table 27 and 28).

Progress of red rot of sugarcane :

Under Hisar conditions during 1989-90, when the canes were inoculated on July 15, resulted in apparantly stationary period upto the September 13, 1989 (Fig. 1). After that there was sudden increase in the expression of red rot incidence and reached upto its maximum extent till October 13, 1989. Inoculations on July 30 and August, 14, 1989 resulted in slow increase in red rot and recorded to the negligible extent in all the three varieties viz., Co 7717, CoL 29 and CoJ 64.

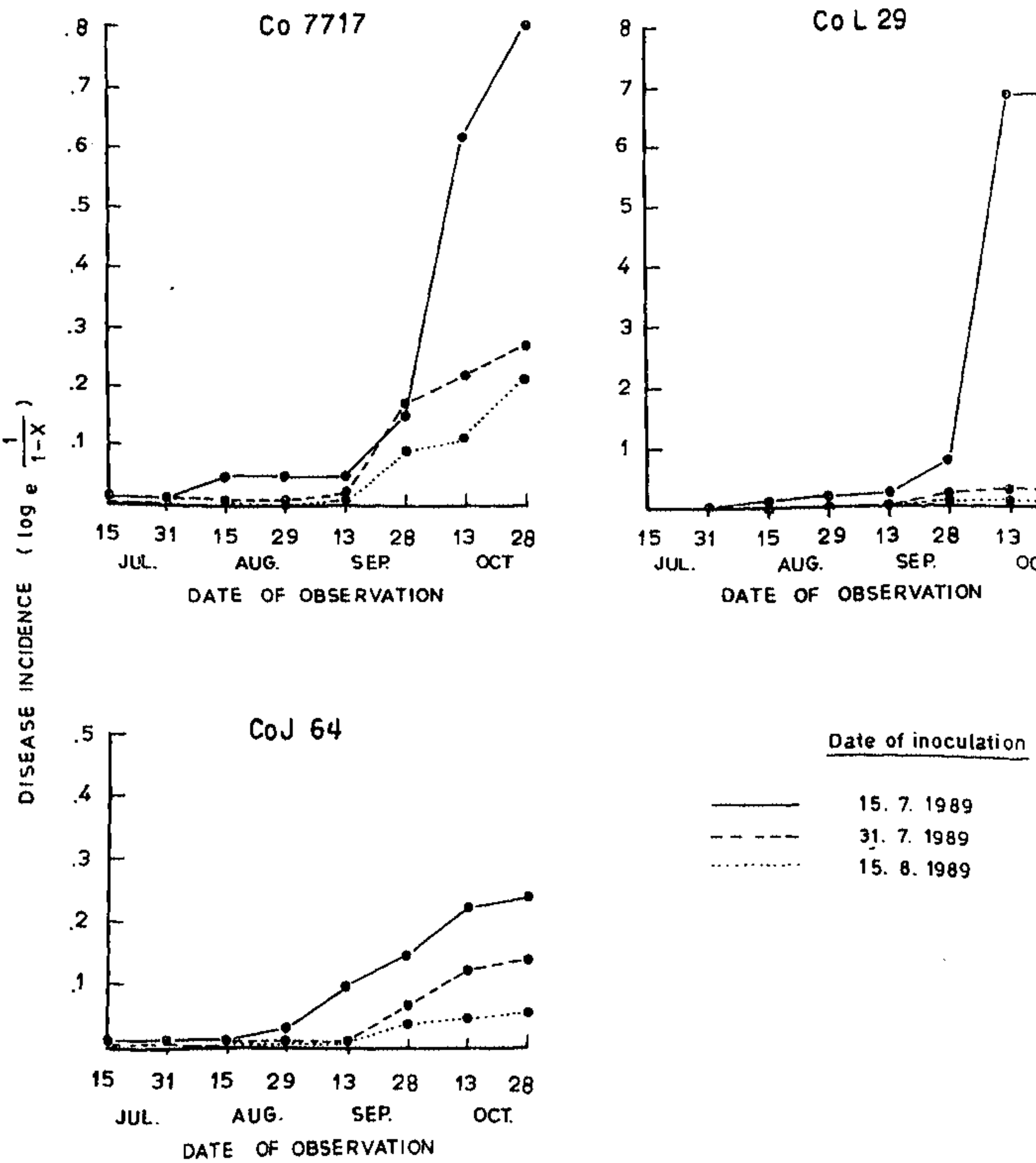

Fig. 1. PROGRESS OF RED ROT OF SUGARCANE (HISAR 1989-90)

Under Uchani (Karnal) conditions 1989-90, when the canes were inoculated on July 20, 1989 resulted in apparently stationary period till August 19, 1989 (Fig. 2). After that there was sudden increase in the expression of red rot and reached upto the maximum till August 20, 1989 whereas when the canes were inoculated on August 4 and 19, 1989, the red rot expression was very slow and recorded to the negligible extent in all the varieties.

Under Hisar condition during 1990-91, when the plants were inoculated on July 3, 1990 (Fig. 3) resulted in apparently stationary period till August 2, 1990. After that there was sudden increase in the expression of red rot incidence and reached to maximum extent till August 17, 1990 whereas when the canes were inoculated on July 18, Augst 2 and August 17, 1990 there was slow increase in the expression of disease.

Under Karnal conditions during 1990-91, when the canes were inoculated on July 5 and 20, 1990 in the variety Co 7717 and only on July 5, 1990 in the varieties CoL 29 and CoJ 64, there was maximum expression of red rot till September 19, 1990 and later periodical inoculation resulted in slow increase in the expression of red rot (Fig. 4).

The accummulated degree days expressed as a cumulative value is used to predict flowering in apple trees to predict the apple scab (Gadoury and Machardy, 1982; James and Sutton, 1982) or even the boll worm (Heliothis spp.) outbreaks and to start insecticidal spray schedule (Wilson and Barnett, 1983).

Karnal bunt caused by Neovossia indica has become a serious disease of wheat in India and causes loss both qualitatively and quantitatively. Following stepwise multiple regression analysis, a linear model to predict the frequency (Y_1)

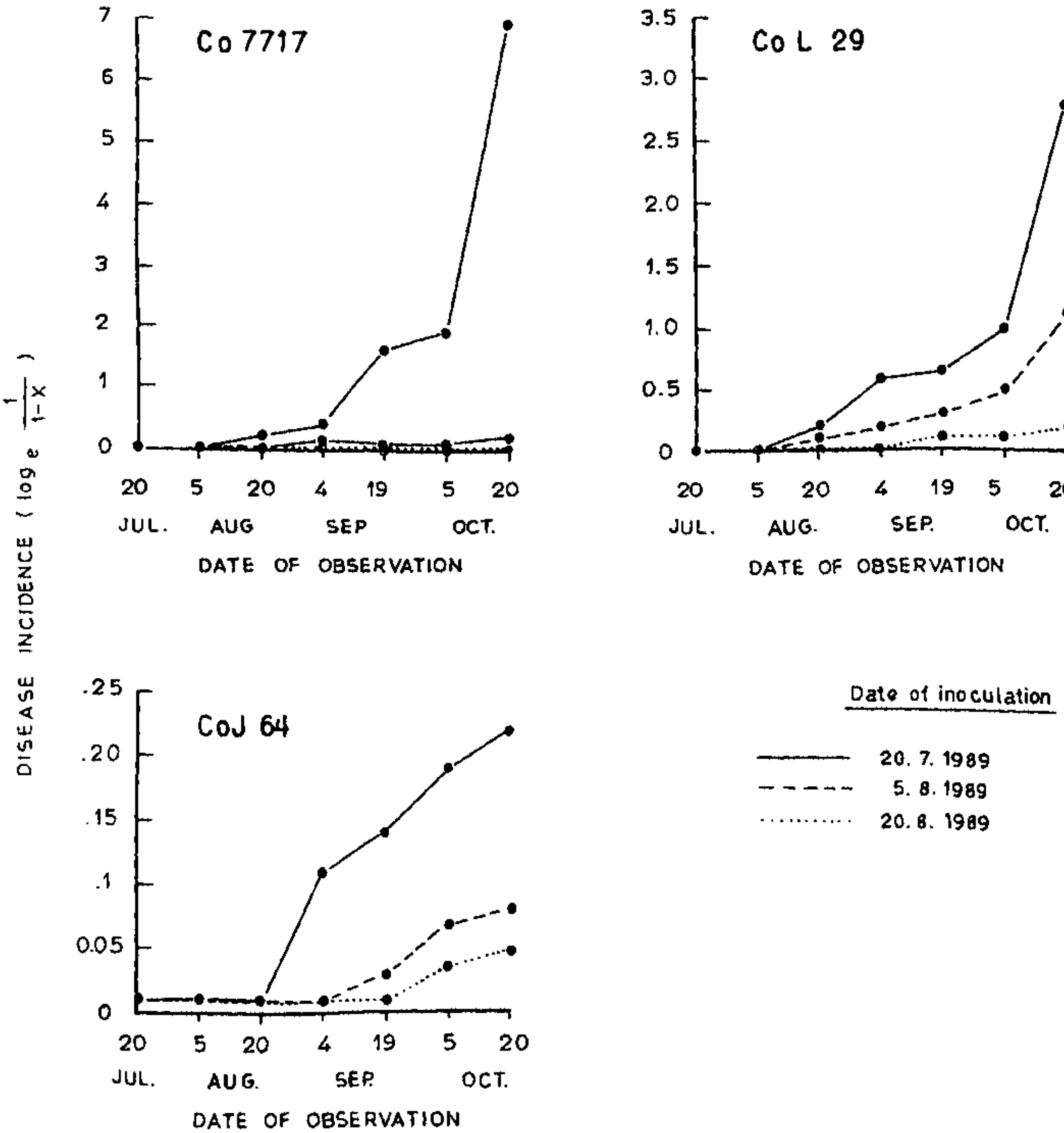

Fig. 2. PROGRESS OF RED ROT OF SUGARCANE (KARNAL 1989-90)

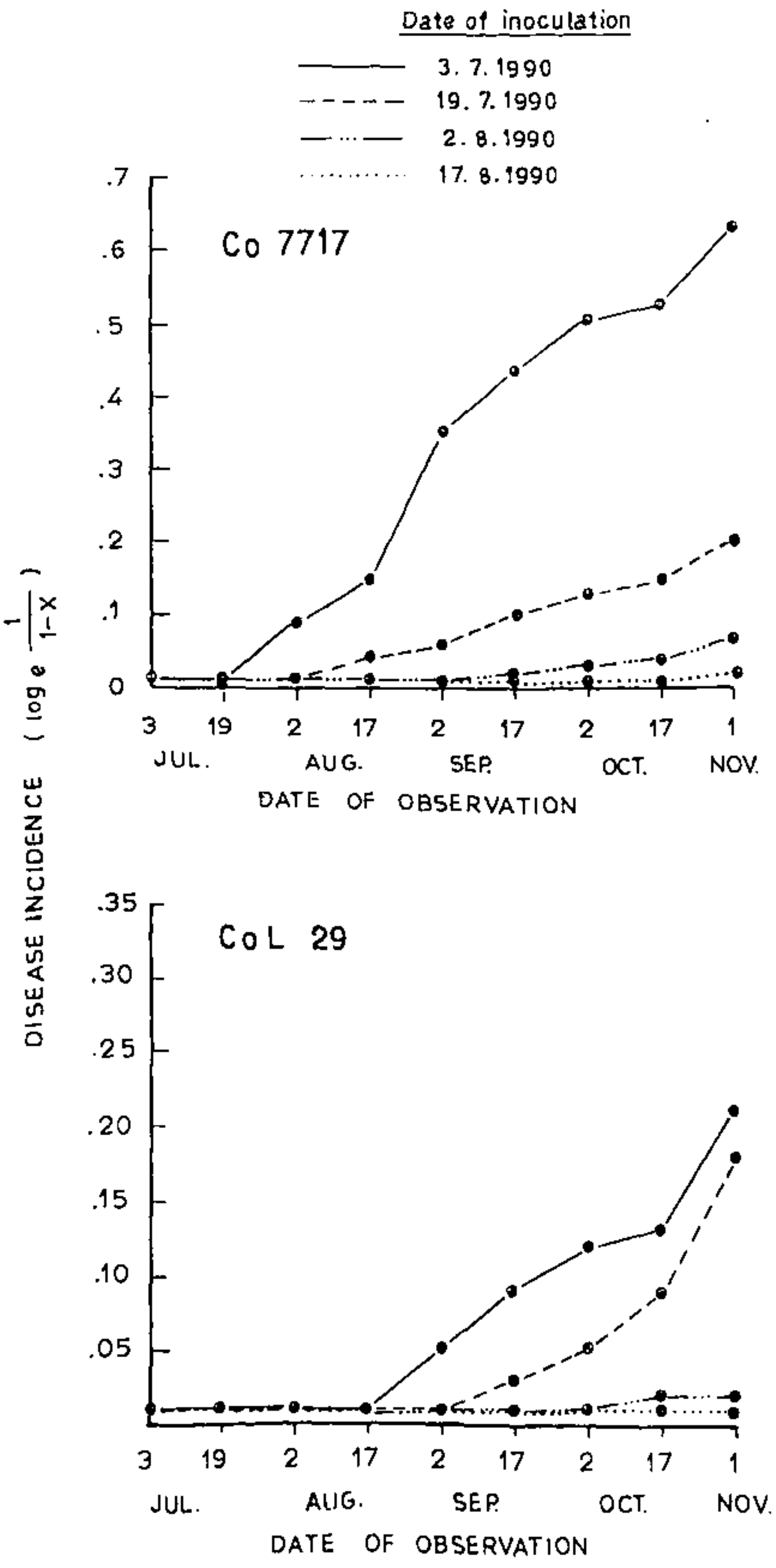

Fig.3 PROGRESS OF RED ROT OF SUGARCANE (HISAR 1990-91)

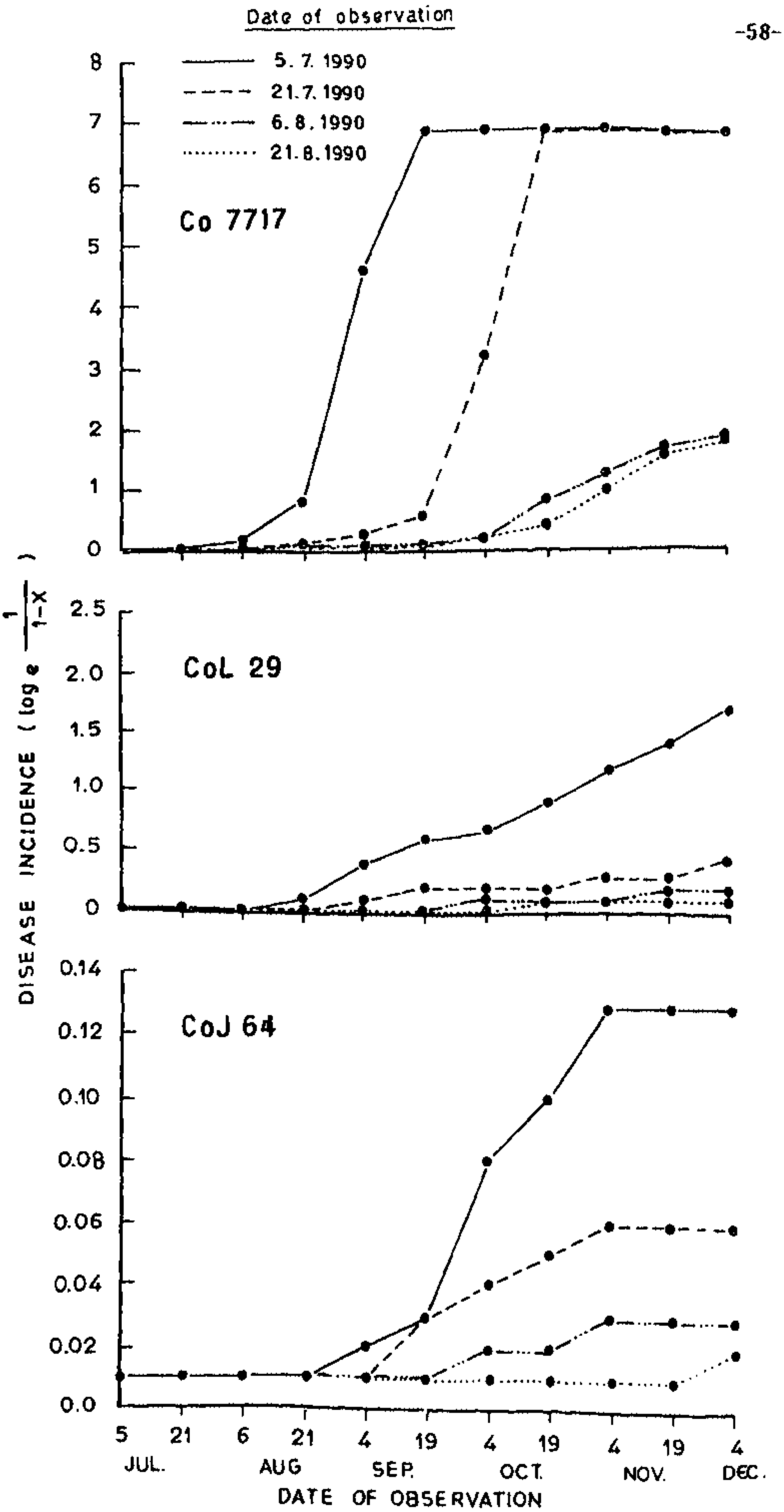

Fig. 4. PROGRESS OF RED ROT OF SUGARCANE(KARNAL 1990-91)

and maximum disease severity (Y_3) of Karnal bunt was developed. A non-linear relationship was observed between temperature and Karnal bunt severity and followed a relationship in second order poly-nominal. It appears that by measuring temperature and rainfall the disease occurrence and severity could be predicted with a reasonable accuracy (Singh et al., 1990).

Only scanty information is available on the role of environmental factors on the development of red rot (Rana and Gupta, 1968; Singh et al., 1984; Singh et al., 1988; Beniwal and Satyavir, 1991). These studies have been conducted using mainly plug method of inoculation. Moreover, in none of the study combined effect of various weather parameters has been worked out and no predictive model has been developed. In the present study an attempt has been made to develop predictive models for red rot using SMRA technique.

From all these analysis and disease progress curves the following points were apparant :

(1) Inoculation of canes on July 3 and 20 during both years at both the locations resulted in better successful infection and expression of red rot.

(2) Inoculation after July 20 resulted in very slow increase in expression of red rot and later on became constant.

(3) Red rot incidence reached to its maximum extent by the end of October and later on expression of red rot became slow and almost constant in all the periodic inoculations.

(4) Red rot expression was negatively correlated with minimum temperature and there was cumulative increase in the expression of red rot. Rate of

increase in the expression of red rot was higher at first decrease in minimum temperature and it decreased with further decrease in minimum temperature because the partial coefficients of minimum temperature was higher at first decrease in minimum temperature and it decreased continuously with further decrease in minimum temperature.

(5) It was the minimum temperature, the most correlated weather parameter followed by relative humidity evening which was negatively correlated during the expression of red rot incidence.

Pathogenic potential of mid-rib isolates :

Ten purified mid-rib isolates of C. falcatum were collected from different sugarcane varieties and localities of Haryana. To study their pathogenic potentials, these isolates alongwith one stalk rot isolate RRK (control) were inoculated by plug method during third week of September, 1990 and observations were recorded after two months.

Linear spread (Table 29) of red rot from mid-rib isolates varied from 5.55 to 24.14 cm, 2.33 to 6.55 cm and 6.47 to 29.30 cm in cultivars Co 7717, CoJ 64 and CoL 29, respectively, While linear spread from stalk rot isolate - RRK was 67.52, 30.35 and 79.33 cm on cultivars Co 7717, CoJ 64 and CoL 29, respectively. Observations recorded indicate that linear spread from mid-ribs isolates was significantly very low as compared to that of the stalk rot isolate-RRK.

Mid-rib isolates demonstrated resistant to moderately reistant reaction to all the three varieties tested whereas moderately susceptible to susceptible reaction was demonstrated by stalk rot isolates (Table 30). Disease rating of all the mid-rib isolates was significantly very low as compared to that of

Table 29 : Linear spread of mid-rib isolates in sugarcane stalks

Mid-rib isolates	Linear Spread (cm) varieties		
	Co 7717	CoJ 64	CoL 29
MR1	5.55	6.55	10.53
MR2	24.14	2.69	26.92
MR3	21.47	3.28	22.53
MR4	12.75	3.78	29.30
MR 5	14.22	6.44	9.47
MR6	17.45	4.75	10.17
MR7	10.97	2.33	11.00
MR8	6.67	3.22	9.31
MR9	6.08	4.64	11.22
MR10	11.14	4.56	6.47
Stalk rot isolate (RRK)	67.52	30.35	79.33
C.D. at 5% level	8.36	3.04	14.84

Table 30 : Disease rating of mid rib isolates in sugarcane stalk

Mid-rib isolates	Disease rating varieties		
	Co 7717	CoJ 64	CoL 29
MR1	0.00(R)	0.61(R)	0.39(R)
MR2	2.22(MR)	0.00(R)	1.64(R)
MR3	2.36(MR)	0.00(R)	2.61(MR)
MR4	1.45(R)	0.00(R)	2.45(MR)
MR5	1.53(R)	0.00(R)	0.67(R)
MR6	1.61(R)	0.22(R)	0.56(R)
MR7	0.67(R)	0.00(R)	0.78(R)
MR8	0.00(R)	0.00(R)	0.33(R)
MR9	0.00(R)	0.00(R)	0.47(R)
MR10	0.61(R)	0.22(R)	0.00(R)
Stalk rot isolate (RRK)	7.39(S)	4.75(MS)	7.78(S)
C.D. at 5% level	0.91	0.54	1.19

R = Resistant
MR = Moderately resistant
S = Susceptible

the stalk rot isolate.

The results of the present study showed that all the mid-rib isolates of C. falcatum under present studies are avirulent on susceptible cultivars Co 7717, CoL 29 and CoJ 64, as such, do not pose any threat to sugarcane cultivation. The present findings do not confirm the earlier findings of Butler (1906), who reported substantial stalk rot after using mid-rib isolates. Rafay and Singh (1957) reported that the light coloured, sporulating mid-rib isolates were virulent, while dark coloured ones were avirulent. No such differentiation was observed in the present study. Agnihotri and Budhraja (1974) reported that most of the mid-rib isolates of C. falcatum studied by them are weak in pathogenic behaviour. The races that cause mid-rib lesions are markedly different from those causing stalk rot. Similarly, Sandhu et al. (1974) have concluded that leaf mid-rib lesion isolates are less virulent than that of stalk isolates.

The results of the present study showed that mid-rib isolates have very little role in pathogenicity of stalk rot. Mid-rib isolates did not cause stalk rot on even the susceptible varieties but there may be the possibility that the virulent type might have evolved from mid-rib isolates by adapting to local host and local climatic conditions (Lewin et al., 1978).

Effect of nitrogen on red rot development :

Experiments were conducted at Hisar and Uchani (Karnal) during 1990-91 to study role of nitrogen on the development of red rot. Three cultivars viz., Co 7717 (highly susceptible), CoJ 64 (susceptible) and CoS 767 (resistant) were planted using randomised block design. Different levels of nitrogen in the form of urea was applied in two split doses. Canes in different treatments were inoculated by plug method and red rot rating was recorded after one month of inoculation.

The data recorded indicate that there was no significant difference in the red rot rating at different levels of nitrogen in varieties Co 7717 and CoS 5767 at both the locations, whereas there was significant increase in red rot rating in cultivar CoJ 64 as the level of nitrogen increased beyond 75 kg/ha at Hisar (Table 31). At Uchani (Karnal) also there was significant increase in red rot rating of cultivar CoJ 64 beyond 75 kg/ha , although red rot rating decreased significantly at the highest dose (300 kg N/ha (Table 32).

There is no information available in the literature regarding role of nitrogen on red rot development. However, higher doses of nitrogen increased the susceptibility of wheat to rust (Puccinia graminis) and powdery mildew (Erysiphe graminae). It may be due to the fact that nitrogen abundance resultedin the production of young, succulent growth and may prolonged the vegetative period and delayed maturity of the plant. These effects made the plant more susceptible to pathogens that normally attack such tissues, and for longer periods (Huber and Watson, 1974). Ramakrishnan (1941) reported that C. falcatum was able to metabolize a number of organic and inorganic sources of nitrogen. Manoch (1963) found that the addition of nitrogen boosted the sucrose invertase production in both the dark and light coloured isolates of red rot pathogen.

In the present study, there was slight increase in the red rot rating at the higher levels of nitrogen application in susceptible cultivar CoJ 64. Cultivar CoJ 64, though susceptible to red rot has been extensively cultivated in Punjab, Haryana and Western Uttar Pradesh for over a decade. The cultivar has tendency to escape red rot under field conditions (Beniwal, 1986)

Table 31 : Effect of nitrogen level on red rot development at Hisar 1990-91

Nitrogen level (Kg/ha)	Disease rating		
	Co 7717	CoJ 64	CoS 767
0	6.53	3.35	0.95
75	6.93	3.85	0.94
150	6.79	4.16	0.85
225	6.58	4.50	0.68
300	6.85	4.83	0.63
C.D. at 5% level	NS	0.77	NS

Table 32 : Effect of nitrogen level on red rot development at Karnal 1990-91

Nitrogen level (Kg/ha)	Disease rating		
	Co 7717	CoJ 64	CoS 767
0	6.64	4.20	2.80
75	6.66	3.63	2.68
150	7.38	5.09	2.68
225	7.94	6.64	2.62
300	7.91	5.43	2.08
C.D. at 5% level	NS	0.80	NS

However, addition of higher nitrogen application might be one of the factors responsible for increase in red rot incidence in this cultivar in North India (Anonymous, 1990). Further detailed investigations are required to confirm the findings.

Effect of different amendments on red rot development :

Amendment of soil is one of the methods of bringing about changes in soil environment regulating the biological balance. The present investigation was conducted to determine the effect of amendments on soil inoculum of C. falcatum under field conditions and their role in red rot development.

Disease incidence on clump and cane basis were recorded in organic and inorganic amended soil (Table 33). There was considerable occurrence of red rot in artificially inoculated soil when healthy setts of cultivar CoL 29 were planted. Similar observations were also reported by Chona (1950). Singh and Singh (1981) reported possible role of survival structures of C. falcatum in soil and infected debries.

The results showed that soil amendment with zinc sulphate @ 15 and 25 kg/ha significantly increased the development of red rot. Possibly it may be due to some stimulatory effect of soil amendment on the activity of the red rot pathogen (Singh, 1985), whereas there was no significant effect on the development of the disease when soil was amended with ferrous sulphate @ 15 and 25 kg/ha.

Soil amendment with sugarcane baggase @ 30 and 40 q/ha and press mud @ 40 and 50 q/ha significantly suppressed the disease incidence which may be due to adverse effect of organic matter on the activity of pathogen or its survival. Decomposition of baggase released some sugars which might

Table 33 : Effect of soil amendments on red rot development

Treatment	Per cent red rot incidence	
	Clump basis	Cane basis
Control	32.90 (35.00)*	16.89 (24.29)
Ferrous sulphate (15 Kg/ha)	35.06 (36.31)	18.39 (25.40)
Ferrous sulphate (25 Kg/ha)	33.33 (35.36)	17.36 (24.63)
Zinc sulphate (15 Kg/ha)	36.53 (37.17)	19.95 (26.52)
Zinc sulphate (25 Kg/ha)	38.29 (38.23)	22.60 (28.37)
Sugarcane baggase (30 q/ha)	25.18 (30.13)	13.68 (21.72)
Sugarcane baggase (40 q/ha)	20.50 (26.91)	12.61 (20.82)
Press mud (40 q/ha)	25.25 (30.16)	12.48 (20.66)
Press mud (50 q/ha)	23.72 (29.14)	13.22 (21.33)
C.D. at 5% level	(1·88)	(1·74)

* Values in parentheses are angular transformed.

have altered the physical, chemical and biotic conditions of the soil (Singh, 1985). He observed manifold increase in microbial populations in soil amended with baggase as com- red to the control. Secondly, due to poor saprophytic ability, C. falcatum can not easily comp ete with enhanced population of antagonists or lytic micro-organisms. Singh (1985) also reported that soil amended with bagasse caused about 25 per cent lysis of microbial growth of C. falcatum, while in check it was only 0.5 per cent. Suppressive effect of press mud on red rot may be attributed to antagonistic micro-organisms or immediate germination of resting structures followed by lysis.

Effect of Bavistin on red rot incidence :

In spite of vigrous seed selection exercised at planting time in respect of elimination of red rot infected setts prepared out of such canes, a few diseased setts can inadvertently escape notice and serve as a source of secondary infection. Therefore, the role of the systemic fungicide Bavistin in controlling the disease was investigated. Healthy setts of highly susceptible cultivar CoL 29 were planted in artifically infested soil with infected plant debries. Setts were treated with Bavistin for one hour and two hour duration alongwith or without one or two sprays of Bavistin. Red rot incidence at cane and clump basis were recorded (Table 34) . Significantly low incidence of red rot was recorded in one and two hour sett treatment alone and alongwith foliar sprays with Bavistin as compared to control. While one and two sprays alone did not reduce the disease incidence significantly.

Lewin et al. (1976) demonstrated good control of the disease even with lower dose (0.25%) and shorter treatment duration (dip and out) with Bavistin. Waraitch (1983) demonstrated that dip and out treatment of red rot

Table 34 : Efficacy of Bavistin against red rot

Treatment	Per cent red rot incidence	
	Clump basis	Cane basis
Control	40.74 (39.69)*	22.97 (28.73)
One hour sett treatment (@ 0.25%)	34.24 (36.19)	19.21 (25.98)
One hour sett treatment + one spray	34.80 (36.11)	16.85 (24.25)
One hour sett treatment + two sprays	32.80 (35.11)	16.57 (23.99)
Two hours sett treatment	32.01 (34.58)	15.60 (23.48)
Two hours sett treatment + one spray	32.60 (34.82)	16.94 (24.26)
Two hours sett treatment + two sprays	34.37 (35.91)	18.55 (25.62)
One spray (@ 0.1%)	38.37 (38.10)	20.98 (27.25)
Two sprays	40.07 (39.47)	21.38 (27.66)
C.D. at 5% level	(1.63)	(2.12)

* Values in parenthesis are the angular transformed.

infected setts with Bavistin did not reduce the disease incidence significantly as compared to control whereas minimum disease incidence was observed in case of one hour sett treatment with Bavistin (0.5%). While reviewing the efficacy of fungicides in controlling red rot, Singh and Singh (1989) reported that fungicidal treatments have not been effective in controlling the disease. Slight reduction in red rot incidence due to Bavistin sett treatments is unlikely to provide effective control under field conditions.

Identification of red rot resistant mutant :

Dormant single budded setts of early maturing high sugared varieties Co 8334 and CoH 56 when subjected to gamma radiation exposure of 4.0 Kilo roentgen (kR) resulted in reduced per cent survival of germinated buds in M_1 generation (Table 35). Canes in all the survival clones in M_1 generation alongwith control were inoculated by nodal method during the month of September, 1989. Seven and three apparently healthy clones from varieties Co-8334 and CoH 56, respectively were selected after visually recording observations on red rot. These clones were subsequently planted alongwith control during 1990-91 at Hisar and inoculated by nodal method with the onset of monsoon (Table 36). All the selected mutants from cultivars Co 8334 and CoH 56 developed red rot during, 1990-91 and none of the mutants showed resistance to red rot in M_2 generation.

A number of earlier workers have identified red rot resistant mutants through gamma irradiations (Rao et al., 1966; Singh, 1970; Shah, 1972; Nair, 1973; Bari, 1974; Haq et al., 1974; Jagathesan , 1974; Jagathesan, et al. ,1977). These workers have used 2-6 KR dose for irradiations. In the present studies irradiation of dormant buds of cultivar Co 8334 and CoH 56 at 4 KR did not produce genetic variants and gamma irradiation was ineffective in inducing red rot resistance.

Table 35 : Identification of red rot resistant mutant

Variety	Gamma rays treatment	No. of buds treated	Per cent survival	No. of mutants selected (M_1)
Co 8334	4 kR	1000	45.90	7
Co 8334	0 kR	0.00	71.25	-
CoH 56	4 kR	1000	65.60	3
CoH 56	0 kR	0.00	35.62	-

Table 36 : Red rot reaction of selected mutants in M_2

	Red rot reaction in M_2 generation
1. Co 8334 -M_1	S
M_2	S
M_3	S
M_4	S
M_5	S
M_6	S
M_7	S
Co 8334 (Control)	S
2. CoH 56 -M_1	S
M_2	S
M_3	S
CoH 56 (Control)	S

S = Susceptible.

Summary

CHAPTER-V

SUMMARY

The main objective of this investigation was to study the epidemiology and management of red rot of sugarcane caused by *Colletotrichum falcatum* Went. Attempts were made to develop the predictive models for the red rot incidence by conducting the experiments at Hisar and Karnal during the years 1989-90 and 1990-91. Standing canes of varieties Co 7717, CoL 29 and CoJ 64 were inoculated by nodal method at 15 days interval starting from the onset of monsoon. Weather data were regressed with disease incidence by applying stepwise multiple regression analysis and best fitting equations having R^2 0.90 are as follows :

Regression equation	R^2
$Y_1 = 53.13 - 1.97X_2$	0.90
$Y_2 = 179.16 - 6.63X_2$	0.97
$Y_1 = 28.55 - 0.32X_4$	0.97
$Y_2 = 24.56 - 0.25X_4$	0.91
$Y_2 = 31.74 - 1.23X_4$	0.96
$Y_1 = 144.70 - 5.35X_2$	0.90
$Y_1 = 148.01 - 5.82X_2$	0.96
$Y_2 = 20.75 - 0.83X_2$	0.90

On the basis of prediction equation and disease progress curves the following points were apparent :

(i) Inoculation of canes on July 3-20 during 1989-90 and 1990-91 at both the locations resulted in successful infection and expression of red rot.

(ii) Inoculations after July 20 resulted in very slow increase in expression of red rot and later became constant.

(iii) Red rot incidence reached to its maximum extent by the end of October and later on, expression of red rot became slow and almost constant in all the periodic inoculations.

(iv) Red rot expression was negatively correlated with minimum temperature (X_2) and there was cumulative increase in the expression of red rot. Rate of increase in the expression of red rot was higher at first decrease in minimum temperature and it decreased with further decrease in minimum temperature because the partial coefficient of minimum temperature was higher at first decrease in minimum temperature and it decreased further with decrease in minimum temperature.

(v) Minimum temperature was the most correlated weather parameter followed by relative humidity evening (X_4) which was negatively correlated during the expression of red rot, where the minimum temperature fails to become significant.

The pathogenic potentials of 10 mid rib isolates, collected from different sugarcane varieties and localities of Haryana alongwith one stalk rot isolates RRK were studied. Observations recorded on linear spread and disease rating showed that all the mid-rib isolates of C. falcatum under present studies are avirulent on susceptible varieties Co 7717, CoL 29 and CoJ 64 and thus, do not pose any threat to sugarcane cultivation.

Field trials with three cultivars Co 7717, CoL 29 and CoJ 64 were conducted at Hisar and Uchani (Karnal) during 1990-91 to study role of nitrogen on the development of red rot. Different levels of nitrogen (0, 75, 150, 225 and 300 kg/ha) were applied in the form of urea. There was no significant difference in the red rot rating at different levels of nitrogen in varieties Co 7717 and CoS 767 at both the locations whereas there was significant increase in red rot rating in variety CoJ 64 with increased nitrogen levels beyond 75kg N/ha.

An experiment was conducted to determine the effect of various amendments on soil inoculum of C. falcatum under field conditions and their role in red rot development. The results showed that soil amendment with zinc sulphate @ 15 and 25 Kg/ha significantly increased the development of red rot whereas soil amendment with sugarcane baggase @ 30 and 40 q/ha and press mud @ 40 and 50 q/ha significantly suppressed the disease incidence.

The role of the systemic fungicide Bavistin in controlling the red rot was investigated. Healthy setts of highly susceptible variety CoL 29 were planted in artifically inoculated soil. Setts were treated with Bavistin for one hour and two hour duration alongwith or without one or two sprays of Bavistin. Slightly low incidence of red rot was recorded in one and two hours setts treatment alone and alongwith foliar sprays with Bavistin as compared to control, while one and two sprays alone did not reduce the disease incidence significantly.

To identify the red rot resistant mutant, dormant signle budded setts of early maturing,high sugared varieties Co 8334 and CoH 56 were subejcted to gamma radiation exposure of 4.0 kR. Gamma radiation resulted in reduced per cent survival of germinated buds in M_1 generation. Canes in all the survival clones in M_1 generation alongwith control were inoculated by nodal method during the month of September, 1989. Healthy clones selected after visually recording observation on red rot were planted alongwith control during 1990-91 at Hisar and inoculated by nodal method with the onset of monsoon. All the selected mutants from varieties Co 8334 and CoH 56 developed red rot during 1990-91 and none of the mutants showed resistant to red rot in M_2 generations.

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* Original not seen.

Details of the sample data from Hisar 1989-90

(Date of inoculation - 15.7.1989)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
15.7.89	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
31.7.89	9.00 (0.010)	0.52 (0.010)	0.00 (0.010)
15.8.89	4.72 (0.048)	10.26 (0.109)	1.25 (0.013)
29.8.89	4.95 (0.051)	13.85 (0.149)	3.25 (0.034)
13.9.89	4.95 (0.051)	19.93 (0.222)	9.52 (0.101)
28.9.89	13.90 (0.151)	53.31 (0.761)	14.26 (0.154)
13.10.89	45.77 (0.609)	100.00 (6.908)	20.15 (0.226)
28.10.89	53.39 (0.764)	100.00 (6.908)	22.14 (0.250)

* Values in parentheses are $\log_e$ (1/1-x) transformed value of disease incidence in per cent.

Details of the sample data from Hisar 1989-90

(Date of inoculation - 13.7.89)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
31.7.89	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
15.8.89	0.00 (0.010)	0.00 (0.010)	0.00 (0.010)
29.8.89	1.09 (0.011)	0.88 (0.010)	0.00 (0.010)
13.9.89	2.31 (0.023)	3.54 (0.036)	1.40 (0.014)
28.9.89	15.24 (0.165)	15.26 (0.166)	6.49 (0.067)
13.10.89	(20.13) (0.224)	26.47 (0.308)	12.26 (0.131)
28.10.89	23.91 (0.273)	28.95 (0.342)	13.57 (0.146)

* Values in parentheses are $\log_e$ (1/1-x) transformed value of disease incidence in per cent.

Details of the sample data from Hisar 1989-90

(Date of inoculation - 15.8.89)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
15.8.89	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
29.8.89	0.00 (0.010)	0.00 (0.010)	0.00 (0.010)
13.9.89	0.75 (0.010)	1.07 (0.010)	0.00 (0.010)
28.9.89	8.66 (0.091)	6.56 (0.068)	3.87 (0.040)
13.10.89	10.67 (0.113)	9.14) (0.095)	4.83 (0.049)
28.10.89	18.67 (0.207)	10.56 (0.112)	5.42 (0.056)

* Values in parentheses are $\log_e$ (1/1-x) transformed value of disease incidence in per cent.

Details of the sample data from Uchani (Karnal) 1989-90

(Date of inoculation - 20.7.89)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
20.7.89	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
5.8.89	0.00 (0.010)	0.00 (0.010)	0.00 (0.010)
20.8.89	18.30 (0.202)	19.59 (0.218)	0.00 (0.010)
4.9.89	31.58 (0.380)	46.81 (0.631)	10.28 (0.109)
19.9.89	78.73 (1.595)	47.48 (0.645)	12.73 (0.136)
5.10.89	85.24 (1.911)	64.19 (1.027)	16.10 (0.176)
20.10.89	100.00 (6.908)	94.04 (2.813)	19.54 (0.216)

* Values in parentheses are $\log_e (1/1-x)$ transformed value of disease incidence in per cent.

Details of the sample data from Uchani (Karnal) 1989-90

(Date of inoculation - 5.8.89)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
5.8.89	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
20.8.89	0.00 (0.010)	5.78 (0.060)	0.00 (0.010)
4.9.89	5.50 (0.057)	17.71 (0.195)	1.06 (0.011)
19.9.89	9.92 (0.099)	28.14 (0.330)	2.77 (0.028)
5.10.89	12.57 (0.135)	38.99 (0.494)	6.28 (0.065)
20.10.89	19.12 (0.212)	67.77 (1.136)	7.76 (0.081)

* Values in parentheses are $\log_e$ (1/1-x) transformed value of disease incidence in per cent.

Details of the sample data from Uchani (Karnal) 1989-90

(Date of inoculation - 20.8.89)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
20.8.89	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
4.9.89	3.07 (0.031)	4.31 (0.043)	0.00 (0.010)
19.9.89	7.38 (0.077)	7.88 (0.082)	1.41 (0.014)
5.10.89	11.72 (0.124)	10.56 (0.112)	3.67 (0.038)
20.10.89	16.73 (0.183)	15.57 (0.170)	5.23 (0.052)

* Values in parentheses are $\log_e$ (1/1-x) transformed value of disease incidence in per cent.

Details of the sample data from Hisar 1990-91

(Date of inoculation - 3.7.1990)

Date of observation	Red rot incidence varieties	
	Co 7717	CoL 29
3.7.90	0.00 (0.010)*	0.00 (0.010)
19.7.90	0.00 (0.010)	0.00 (0.010)
2.8.90	8.19 (0.086)	0.00 (0.010)
17.8.90	13.43 (0.145)	0.65 (0.010)
2.9.90	29.48 (0.348)	5.24 (0.053)
17.9.90	35.01 (0.431)	8.53 (0.089)
2.10.90	39.50 (0.503)	11.27 (0.120)
17.10.90	40.42 (0.518)	11.80 (0.216)
1.11.90	46.63 (0.627)	19.81 (0.221)

* Values in parentheses are $\log_e$ (1/1-x) transformed value of disease incidence in per cent.

Details of the sample data from Hisar 1990-91

(Date of inoculation - 19.7.1990)

Date of observation	Red rot incidence varieties	
	Co 7717	CoL 29
19.7.90	0.00 (0.010)*	0.00 (0.010)
2.8.90	0.00 (0.010)	0.00 (0.010)
17.8.90	3.84 (0.039)	0.00 (0.010)
2.9.90	6.19 (0.064)	1.00 (0.010)
17.9.90	9.14 (0.104)	2.62 (0.026)
2.10.90	11.77 (0.126)	4.78 (0.049)
17.10.90	14.10 (0.152)	8.49 (0.089)
1.11.90	20.78 (0.233)	16.60 (0.182

* Values in parentheses are log $_e$ (1/1-x) transformed value of disease incidence in per cent.

Details of the sample data from Hisar 1990-91

(Date of inoculation - 2.8.1990)

Date of observation	Red rot incidence varieties	
	Co 7717	CoL 29
2.8.90	0.00 (0.010)*	0.00 (0.010)
17.8.90	0.65 (0.010)	0.00 0.010)
2.9.90	1.28 (0.013)	0.00 (0.010)
17.9.90	2.38 (0.024)	0.00 (0.010)
2.10.90	3.00 (0.030)	1.30 (0.013)
17.10.90	4.01 (0.041)	1.56 (0.016)
1.11.90	6.28 (0.065)	2.00 (0.020)

* Values in parentheses are $\log_e (1/1-x)$ transformed value of disease incidence in per cent.

Details of the sample data from Hisar 1990-91

(Date of inoculation - 17.8.1990)

Date of observation	Red rot incidence varieties	
	Co 7717	CoL 29
17.8.90	0.00 (0.010)*	0.00 (0.010)
2.9.90	0.00 (0.010)	0.00 (0.010)
17.9.90	0.39 (0.010)	0.00 (0.010)
2.10.90	0.75 (0.010)	0.00 (0.010)
17.10.90	1.00 (0.010)	0.66 (0.010)
1.11.90	2.50 (0.025)	1.00 (0.010)

* Values in parentheses are $\log_e$ (1/1-x) transformed value of disease incidence in per cent.

Appendix 4.1

Details of the sample data from Uchani (Karnal) 1990-91

(Date of inoculation - 5.7.1990)

Date of Observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
5.7.90	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
21.7.90	9.09 (0.095)	0.00 (0.010)	0.00 (0.010)
6.8.90	54.92 (0.796)	2.27 (0.023)	0.00 (0.010)
21.8.90	99.07 (4.605)	8.07 (0.084)	1.19 (0.012)
5.9.90	100.00 (6.908)	33.48 (0.408)	2.38 (0.024)
20.9.90	100.00 (6.908)	43.13 (0.565)	3.31 (0.033)
5.10.90	100.00 (6.908)	51.67 (0.728)	7.45 (0.078)
20.10.90	100.00 (6.908)	60.83 (0.937)	10.39 (0.110)
5.11.90	100.00 (6.908)	68.88 (1.168)	11.82 (0.126)
20.11.90	100.00 (6.908)	75.85 (1.423)	12.15 (0.130)
4.12.90	100.00 (6.908)	82.80 (1.720)	12.15 (0.130)

* Values in parentheses are $\log_e(1/1-x)$ transformed value of disease incidence in per cent.

Appendix 4.2

Details of the sample data from Uchani (Karnal) 1990-91

(Date of inoculation - 21.7.1990)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
21.7.90	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
6.8.90	1.50 (0.104)	0.00 (0.010)	0.00 (0.010)
21.8.90	9.88 (0.278)	1.08 (0.011)	0.00 (0.010)
5.9.90	23.80 (0.620)	11.26 (0.120)	0.00 (0.010)
20.9.90	96.45 (3.219)	13.71 (0.147)	0.00 (0.010)
5.10.90	100.00 (6.908)	16.17 (0.177)	3.55 (0.020)
20.10.90	100.00 (6.908)	21.26 (0.240)	4.99 0.020)
5.11.90	100.00 (6.908)	24.94 (0.286)	5.66 (0.027)
20.11.90	100.00 (6.908)	29.03 (0.342)	5.66 (0.034)
4.12.90	100.00 (6.908)	30.64 (0.365)	5.66 (0.034)

* Values in parentheses are $\log_e (1/1-x)$ transformed value of disease incidence in per cent.

Appendix 4.3

Details of the sample data from Uchani(Karnal) 1990-91

(Date of inoculation - 6.8.1990)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
6.8.90	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
21.8.90	0.00 (0.010)	0.00 (0.010)	0.00 (0.010)
5.9.90	9.18 (0.097)	2.67 (0.027)	0.00 (0.010)
20.9.90	11.81 (0.126)	4.33 (0.044)	0.00 (0.010)
5.10.90	21.32 (0.240)	6.67 (0.069)	2.10 (0.020)
20.10.90	60.90 (0.939)	9.33 (0.098)	2.10 (0.020)
5.11.90	72.70 (1.284)	12.51 (0.134)	2.67 (0.027)
20.11.90	81.22 (1.671)	16.00 (0.174)	3.33 (0.034)
4.12.90	84.73 (1.877)	16.67 (0.183)	3.33 (0.034)

* Values in parentheses are $\log_e (1/1-x)$ transformed value of disease incidence in per cent.

Appendix 4.4

Details of the sample data from Uchani (Karnal) 1990-91

(Date of inoculation - 21.8.1990)

Date of observation	Red rot incidence varieties		
	Co 7717	CoL 29	CoJ 64
21.8.90	0.00 (0.010)*	0.00 (0.010)	0.00 (0.010)
5.9.90	0.00 (0.010)	0.00 (0.010)	0.00 (0.010)
20.9.90	6.83 (0.070)	0.50 (0.010)	0.00 (0.010)
5.10.90	14.96 (0.163)	1.04 (0.010)	0.53 (0.010)
20.10.90	35.20 (0.434)	6.12 (0.063)	1.32 (0.013)
5.11.90	61.26 (0.945)	8.22 (0.086)	1.32 (0.013)
20.11.90	79.82 (1.599)	11.72 (0.124)	1.32 (0.013)
4.12.90	95.33 (3.058)	12.67 (0.136)	1.94 (0.019)

* Values in parentheses are $\log_e (1/1-x)$ transformed value of disease incidence in per cent.

INFORMATIONS OF WEATHER PARAMETERS

Hisar 1989-90

Period	X_1	X_2	X_3	X_4	X_5
15.7.89 - 30.7.89	36.9	26.5	75.4	39.3	0.23
31.7.89 - 14.8.89	37.1	25.5	73.5	40.1	0.10
15.8.89 - 29.8.89	34.0	24.9	87.5	62.9	5.10
30.8.89 - 13.9.89	35.2	22.7	76.3	43.3	0.33
14.9.89 - 28.9.89	36.0	22.4	78.2	39.0	0.27
29.9.89 - 13.10.89	36.3	18.4	72.7	23.5	0.00
14.10.89 - 28.10.89	33.4	13.4	77.1	22.1	0.00

Karnal 1989-90

Period	X_1	X_2	X_3	X_4	X_5
20.7.89 - 4.8.89	33.1	25.6	85.3	69.0	5.43
5.8.89 - 19.8.89	33.4	25.7	91.1	76.6	2.13
20.8.89 - 4.9.89	30.7	24.3	93.9	82.1	10.00
5.9.89 - 19.9.89	33.8	23.7	87.7	62.9	0.00
20.9.89 - 5.10.89	33.5	20.6	88.9	52.2	0.00
6.10.89 - 20.10.89	33.9	16.9	79.0	38.3	0.00

Contd........

Hisar 1990-91

Period	X_1	X_2	X_3	X_4	X_5
3.7.90 - 18.7.90	33.8	26.1	83.8	61.0	5.55
19.7.90 - 2.8.90	35.1	25.4	85.7	66.7	4.38
3.8.90 -17.8.90	34.1	25.3	89.1	63.8	3.36
18.8.90 - 2.9.90	35.8	25.5	84.3	54.4	0.14
3.9.90 -17.9.90	33.6	24.2	86.0	63.9	10.45
18.9.90 - 2.10.90	33.6	22.4	83.3	54.4	0.71
3.10.90 -17.10.90	33.4	18.5	85.7	39.7	0.00
18.10.90 - 1.11.90	30.8	12.4	78.2	25.0	0.00

Karnal 1990-91

Period	X_1	X_2	X_3	X_4	X_5
5.7.90 -20.7.90	32.6	25.8	83.2	78.8	4.21
21.7.90 - 5.8.90	32.9	25.7	90.2	75.9	15.34
6.8.90 -20.8.90	31.9	25.6	87.5	82.2	2.89
21.8.90 - 4.9.90	33.1	25.5	86.6	73.7	8.10
5.9.90 -19.9.90	31.4	24.6	91.3	78.2	9.83
20.9.90 - 4.10.90	32.4	22.4	86.8	61.4	1.15
5.10.90 -19.10.90	30.7	18.8	83.3	57.5	1.87
20.10.90 - 4.11.90	29.9	13.7	79.3	35.2	0.00
5.11.90 -19.11.90	29.7	12.6	86.3	35.3	0.00
20.11.90 -4.12.90	24.8	9.3	84.5	54.3	1.61

Hisar 1990-91

Period	X_1	X_2	X_3	X_4	X_5
3.7.90 - 18.7.90	33.8	26.1	83.8	61.0	5.55
19.7.90 - 2.8.90	35.1	25.4	85.7	66.7	4.38
3.8.90 -17.8.90	34.1	25.3	89.1	63.8	3.36
18.8.90 - 2.9.90	35.8	25.5	84.3	54.4	0.14
3.9.90 -17.9.90	33.6	24.2	86.0	63.9	10.45
18.9.90 - 2.10.90	33.6	22.4	83.3	54.4	0.71
3.10.90 -17.10.90	33.4	18.5	85.7	39.7	0.00
18.10.90 - 1.11.90	30.8	12.4	78.2	25.0	0.00

Karnal 1990-91

Period	X_1	X_2	X_3	X_4	X_5
5.7.90 -20.7.90	32.6	25.8	83.2	78.8	4.21
21.7.90 - 5.8.90	32.9	25.7	90.2	75.9	15.34
6.8.90 -20.8.90	31.9	25.6	87.5	82.2	2.89
21.8.90 - 4.9.90	33.1	25.5	86.6	73.7	8.10
5.9.90 -19.9.90	31.4	24.6	91.3	78.2	9.83
20.9.90 - 4.10.90	32.4	22.4	86.8	61.4	1.15
5.10.90 -19.10.90	30.7	18.8	83.3	57.5	1.87
20.10.90 - 4.11.90	29.9	13.7	79.3	35.2	0.00
5.11.90 -19.11.90	29.7	12.6	86.3	35.3	0.00
20.11.90 -4.12.90	24.8	9.3	84.5	54.3	1.61